Uniform deviation inequalities in statistical learning

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We first recall some standard deviation inequalities such as Chernoff multiplicative and sub-Gaussian. We then introduce some useful concepts to deal with suprema of empirical processes such as symmetrization and the Rademacher complexity. We then present two Vapnik type inequalities: the original one from Vapnik's seminal work and one other with relative deviation that improved the original one. In addition, a Bernstein-type (with the variance) deviation inequality for suprema of empirical processes is established for general classes that have a reasonably large covering number (highlighting a certain control on the complexity of the classes). The proposed approach turns out to be useful in many statistical learning application including: (i) *local averaging* estimators for which "fast learning rates" are obtained as well as uniform error bounds with respect to the query point; (ii) *empirical risk minimization* estimators.

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1 Motivation

The aim of the course is to obtain probability bound, i.e., $\mathbb{P}(|Z| > t) \leq \delta$, for certain quantity of interest Z that results from Machine learning algorithm or statistical methods. One first estimator that will be study in this lecture notes is the kernel density estimator (KDE) which is now introduced under the following framework. Let $(X, X_1, \ldots, X_n)$ be a sequence of independent and identically distributed random elements with common distribution P_X on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. We suppose that P_X has a density with respect to Lebesgue measure, i.e., for all $B \in \mathcal{B}(\mathbb{R}^d)$, $P_X(B) = \int_B f_X \mathrm{d}\lambda$ where f_X is called the density function of X. The KDE density estimate is given by

$$\hat{f}_n(x) = n^{-1} \sum_{i=1}^{n} K_h(x - X_i), \qquad x \in \mathbb{R}^d,$$

where K is a probability density function and K_h is the rescaled version so that it has standard deviation hI_d. Now assuming that f_X is L_X-Lipschitz, one might show easily that

$$|\mathbb{E}[\hat{f}_n(x)] - f_X(x)| \leq hL_X$$

and, for h small enough,

$$\mathrm{var}(\hat{f}_n(x)) \leq \frac{2f(x)}{nh^d}.$$

This advocates for an error decomposition of the form

$$|\hat{f}_n(x) - f(x)| \leq hL_X + \sqrt{\frac{2f(x)}{nh^d} \log(1/\delta)}$$

The previous holding true with probability $1 - \delta$. This is left as an exercise in the next section, where suitable inequalities will be given. The previous upper bound is non-asymptotic, meaning that it is valid for all $n \geq 1$ which, compared to an asymptotic bound, is more relevant for applications because the upper bound is more precise. However, the bound is only valid at a given x and we have no idea of how the upper-bound evolve when the error is measured over an interval or the full domain, such as

$$\sup_{x \in \mathbb{R}} |\hat{f}_n(x) - f(x)|.$$

The tools developed in this note will allow to obtain such a bound. Application of interest includes Nadaraya-Watson estimator which is a kernel based estimator of the regression function. Another estimator of interest is the popular k-NN regression or classification rule as well as regression tree estimator. The last estimator are well-known instances of a broad framework known as local averaging rule because they all use an averaging over small parts of the data (localized in space) to make the prediction. Other types of estimator that will be covered include empirical risk minimizer (ERM). In that case, one wishes to show that $P\ell(Z, \hat{h}_n) \leq \min_{h \in \mathcal{H}} P\ell(Z, h) + r_n$, where $\hat{h}_n$ is defined as one minimum argument of an empirical estimate of $P\ell(Z, h)$ over $\mathcal{H}$. To study such kind of estimator and obtain a non-asymptotic description of r_n, we will need more than point-wise error but uniform error on the estimator of the risk function. In all previous cases, one major issue is to be able to take advantage of small variance.

2 Basic concentration inequalities

The aim here is to recall some standard background on concentration inequalities. All stated results will be useful for the course. Most of them follow from the Chernoff bound which for any real-valued random variable asserts that (for the right tail), for all $t \in \mathbb{R}$,

$$\mathbb{P}(Z > t) \leq \inf_{\lambda > 0} \mathbb{E}[\exp(\lambda Z)] \exp(-\lambda t)$$

A similar bound can be obtained for the left tail. The moment generating function (or Laplace transform) therefore plays an important role when applying the Chernoff method.

2.1 Multiplicative Chernoff bound

The following result, due to Hagerup and Rüb (1990), is known as the multiplicative Chernoff bound. It is based on an upper bound of the Laplace transform of Bernoulli random variables.

Theorem 1. *Let $(Z_i)_{i=1,\dots,n}$ be a collection of independent random variables with common distribution $\mathcal{B}(\mu)$, $\mu \in (0,1)$. For any $\eta > 0$ and $n \geq 1$, we have*

$$\mathbb{P}\left(\sum_{i=1}^{n} Z_i < (1-\eta)n\mu\right) \leq \exp(-\eta^2 n\mu/2),$$
$$\mathbb{P}\left(\sum_{i=1}^{n} Z_i > (1+\eta)n\mu\right) \leq \exp(-(\eta \wedge \eta^2)n\mu/3).$$

Proof. Consider the upper bound (second inequality). Let $Z \sim \mathcal{B}(\mu)$. The Laplace transform satisfies

$$\mathbb{E}[\exp(\lambda Z)] = \mu \exp(\lambda) + (1-\mu) = 1 + \mu(\exp(\lambda) - 1) \leq \exp(\mu(\exp(\lambda) - 1))$$

and optimizing with respect to λ the quantity $\exp(\mu(\exp(\lambda) - 1) - t\lambda)$ minimum value is $\exp(t - \mu + t\log(\mu/t))$. Setting $t = \mu(1+\eta)$, the Chernoff bound implies that

$$\mathbb{P}(Z > t) \leq (\exp(\eta - (1+\eta)\log(1+\eta)))^{\mu}.$$

One can use the following inequality, for all $\eta \in (0,1]$,

$$\eta - (1+\eta)\log(\eta+1) \leq -(1/3)\eta^2$$

which can be obtained from

$$\eta - (1+\eta)\log(\eta+1) = \int_0^{\eta} \frac{u-\eta}{1+u} du = -\frac{1}{2}\eta^2 + \frac{1}{2}\int_0^{\eta} \frac{(u-\eta)^2}{(1+u)^2} du,$$

and using that $1 + u \geq 1$, we get $\int_0^{\eta} \frac{(u-\eta)^2}{(1+u)^2} du \leq \eta^3/3$, hence

$$\eta - (1+\eta)\log(1+\eta) \leq -\frac{\eta^2}{2} + \frac{\eta^3}{6} \leq -\frac{\eta^2}{3} \qquad (\eta \in (0,1]).$$

We get that, for $\eta \in (0,1]$,

$$\mathbb{P}(Z > \mu(1+\eta)) \leq \exp(-(1/3)\mu\eta^2)$$

The result can easily be extended to sums of independent Bernoulli random variables and we obtain the second inequality. The lower bound (first inequality) can be obtained similarly but using

$$\mathbb{P}(Z < t) \le \mathbb{E}[\exp(-\lambda Z + \lambda t)] \le \exp(\mu(\exp(-\lambda) - 1)) \exp(\lambda t),$$

and using that $\eta + (1-\eta)\log(1-\eta) \ge \frac{1}{2}\eta^2$ when $\eta \in (0, 1]$.

In case $\eta > 1$, the lower bound (first inequality) is still valid while the upper bound needs to be modified into the following sub-exponential bound: for all $\eta > 1$,

$$\mathbb{P}\left(\sum_{i=1}^{n} Z_i > (1+\eta)n\mu\right) \le \exp(-\eta n\mu/3).$$

This is because for all $\eta \ge 1$, $\eta - (1+\eta)\log(1+\eta) \le -\eta/3$. Set $g(\eta) := (1+\eta)\log(1+\eta) - \eta - \eta/3$ on $[1, \infty)$. Then $g'(\eta) = \log(1+\eta) - 1/3 \ge \log 2 - 1/3 > 0$ for $\eta \ge 1$, so g is increasing on $[1, \infty)$ and it suffices to check $g(1) = 2\log 2 - \frac{4}{3} > 0$. Merging the two bounds $\eta \in (0, 1]$ and $\eta > 1$ leads to the stated inequality. □

To compare several tail deviations, one way is to fix the probability to $1 - \delta$. For instance, set $x := 3\log(1/\delta)/(n\mu)$ and $\eta := x \vee \sqrt{x}$. If $x \le 1$ then $\eta = \sqrt{x} \le 1$, and if $x > 1$ then $\eta = x > 1$; in both cases $\eta \wedge \eta^2 = x$. The upper bound then gives $\mathbb{P}\big(\sum_{i=1}^{n} Z_i > (1+\eta)n\mu\big) \le \exp(-x\, n\mu/3) = \delta$, that is, with probability at least $1 - \delta$,

$$\sum_{i=1}^{n}(Z_i - \mu) \le 3\log(1/\delta) \vee \sqrt{3n\mu\log(1/\delta)}.$$

For the lower bound, one may choose $\eta = \sqrt{2\log(1/\delta)/(n\mu)}$ and we obtain that, with probability at least $1 - \delta$,

$$\sum_{i=1}^{n}(Z_i - \mu) \ge -\sqrt{2n\mu\log(1/\delta)}.$$

Putting together the two previous inequalities and using the union bound, we obtain the following two-sided bound: with probability $1 - 2\delta$,

$$\left|\sum_{i=1}^{n}(Z_i - \mu)\right| \le \sqrt{3n\mu\log(1/\delta)} + 3\log(1/\delta).$$

Note finally that when applying Theorem 1 to $1 - Z_i$ instead of Z_i one could obtain a new upper and a new lower bound that might be advantageous compared to the initial ones depending on the value of μ (see Exercise 1 for another upper bound).

Theorem 2. *Let $Z_1, \dots, Z_n$ be independent with $Z_i \sim \mathcal{B}(\mu_i)$, $\mu_i \in [0, 1]$. Then*

$$\mathbb{P}\left(\sum_{i=1}^{n} Z_i < (1-\eta)\sum_{i=1}^{n}\mu_i\right) \le \exp(-\eta^2 \sum_{i=1}^{n}\mu_i/2), \qquad \eta > 0,$$

$$\mathbb{P}\left(\sum_{i=1}^{n} Z_i > (1+\eta)\sum_{i=1}^{n}\mu_i\right) \le \exp(-(\eta^2 \wedge \eta)\sum_{i=1}^{n}\mu_i/3), \qquad \eta > 0$$

Proof. Set $S = \sum_{i=1}^n Z_i$, $m = \mathbb{E}S = \sum_{i=1}^n \mu_i$. For every $\lambda \in \mathbb{R}$, by independence and $1 + u \le e^u$,

$$\mathbb{E}e^{\lambda S} = \prod_{i=1}^n \big(1 + \mu_i(e^\lambda - 1)\big) \le \exp\Big(\sum_{i=1}^n \mu_i(e^\lambda - 1)\Big) = \exp\big(m(e^\lambda - 1)\big).$$

This is the bound used in the i.i.d. case with $n\mu$ replaced by m (valid for both signs of λ, since $1 + u \le e^u$ for all real u, and $1 + \mu_i(e^\lambda - 1) \ge 0$).

Let $\psi(\eta) = (1+\eta)\log(1+\eta) - \eta$. By Markov's inequality and the moment generating function bound, for $\lambda > 0$,

$$\mathbb{P}(S > (1+\eta)m) \le \exp\big(m(e^\lambda - 1) - \lambda(1+\eta)m\big),$$

and the choice $\lambda = \log(1+\eta)$ gives $\mathbb{P}(S > (1+\eta)m) \le e^{-m\psi(\eta)}$. Since $\psi(\eta) \ge \eta^2/3$ on $(0,1]$ and $\psi(\eta) \ge \eta/3$ on $[1,\infty)$ (see above), $\psi(\eta) \ge (\eta^2 \wedge \eta)/3$. For the lower tail and $\eta \in (0,1)$, take $\lambda = -\log(1-\eta)$ in $\mathbb{P}(S < (1-\eta)m) \le \exp\big(m(e^{-\lambda} - 1) + \lambda(1-\eta)m\big)$ and conclude as in the i.i.d. case; for $\eta \ge 1$ the event is empty. The case $m = 0$ is trivial. □

2.2 Sub-Gaussian bound

Let us now introduce sub-Gaussian random variables.

Definition 3. *Let Z be a real-valued random variable. It is called sub-Gaussian with (variance) factor $v > 0$ if*

$$\mathbb{E}[Z] = 0, \qquad \mathbb{E}[\exp(\lambda Z)] \le \exp(\lambda^2 v/2), \quad \forall \lambda \in \mathbb{R}.$$

The factor $v > 0$ is often called variance factor because when Z is sub-Gaussian with factor v then $\mathrm{var}(Z) \le v$ (see Exercise 2). Note that any centered Gaussian random variable Z is sub-Gaussian with factor $v = \mathrm{var}(Z)$. Interestingly, sub-Gaussian is stable under convolution: summing independent sub-Gaussian random variables is still sub-Gaussian. As a consequence, we obtain the following bound for the sum of independent sub-Gaussian random variables.

Theorem 4. *Let $(Z_i)_{i=1,\dots,n}$ be a collection of independent sub-Gaussian random variables such that each $Z_i - \mathbb{E}[Z_i]$ is sub-Gaussian with factor v_i. Then, for any $n \ge 1$ and $t > 0$, we have*

$$\mathbb{P}\left(\sum_{i=1}^n (Z_i - \mathbb{E}Z_i) > t\right) \le \exp\left(-\frac{t^2}{2\sum_{i=1}^n v_i}\right).$$

Proof. Set $W = \sum_{i=1}^n (Z_i - \mathbb{E}Z_i)$ and $V = \sum_{i=1}^n v_i$. By independence, for all $\lambda \in \mathbb{R}$, $\mathbb{E}e^{\lambda W} = \prod_i \mathbb{E}e^{\lambda(Z_i - \mathbb{E}Z_i)} \le e^{\lambda^2 V/2}$, so W is sub-Gaussian with factor V. By Markov's inequality, for $\lambda > 0$, $\mathbb{P}(W > t) \le \exp(\lambda^2 V/2 - \lambda t)$, and $\lambda = t/V$ gives $\exp(-t^2/(2V))$. □

Note that the same holds true for the left-tail because $-Z$ is still sub-Gaussian. Important examples of sub-Gaussian random variables are Gaussian random variables and bounded random variables. In particular, one may show that whenever $Z \in [a,b]$, $Z - \mathbb{E}Z$ is sub-Gaussian with factor $(b-a)^2/4$. This is the subject of Exercise 3. Applying the previous to $Z_i \in [a_i, b_i]$ we obtain the following bound, which is known as the Hoeffding bound and might be seen as an additive version of the previous multiplicative bound.

Theorem 5. *Let $(Z_i)_{i=1,\dots,n}$ be an independent collection of random variables such that each Z_i is valued in $[a_i, b_i]$, where $a_i \leq b_i$ and $\sum_{i=1}^n (b_i - a_i)^2 > 0$. Then, for any $n \geq 1$ and $t > 0$, we have*

$$\mathbb{P}\left(\sum_{i=1}^n \{Z_i - E[Z_i]\} > t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Proof. After checking that each $Z_i - \mathbb{E}Z_i$ is sub-Gaussian with factor $(b_i - a_i)^2/4$. The proof follows from applying Theorem 4. □

From the above, we can deduce the following. For any $n \geq 1$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left|\sum_{i=1}^n \{Z_i - \mathbb{E}[Z_i]\}\right| \leq \sqrt{\frac{\sum_{i=1}^n (b_i - a_i)^2}{2} \log\left(\frac{2}{\delta}\right)}.$$

Note that, in contrast with the multiplicative bound given in Theorem 1, the variance $\mathrm{var}(Z_i)$ is not involved in the Hoeffding bound. This results in a poor scaling when the variance $\mathrm{var}(Z_i)$ is small compared to $(b_i - a_i)^2/4$. This is the case for Bernoulli random variable $\mathcal{B}(\mu)$ with parameter μ that is small (see Exercise 4).

Let us remark that, if η is a Rademacher random variable, i.e., $P(\eta = 1) = P(\eta = -1) = 1/2$ and g is any number, the random variable ηg is sub-Gaussian with variance factor g^2. This will be useful when examining the Rademacher complexity of certain empirical processes (see Section 3). More generally, if Z is a sub-Gaussian random variable with parameter v and $a \in \mathbb{R}$, then aZ is sub-Gaussian with parameter $a^2 v$. This will be useful in our application to k-NN regression in the next section.

As underlined in Exercise 4, the Chernoff multiplicative bound might provide more accurate estimate than the sub-Gaussian approach, especially when the sub-Gaussian factor is loose. For instance the sub-Gaussian applied to KDE with sub-Gaussian factor $1/h^{2d}$ (this is called the Hoeffding inequality) leads to a convergence rate of order $1/\sqrt{n}h^d + h$, that is obviously sub-optimal given that consistency results on KDE are obtained under the weaker condition that $nh^d \to +\infty$. This however would be obtained by relying Chernoff multiplicative bound (see Exercise 8).

One other interesting inequality that is relevant in many situations is Bernstein inequality

Theorem 6 (Bernstein's inequality)**.** *Let $Z, Z_1, \dots, Z_n$ be independent and identically distributed random variables such that $\sigma^2 = \mathrm{var}(Z)$ and $|Z - \mathbb{E}[Z]| \leq b$ almost surely. Then, for all $t > 0$,*

$$\mathbb{P}\left(\sum_{i=1}^n (Z_i - \mathbb{E}Z) > t\right) \leq \exp\left(-\frac{t^2}{2\left(n\sigma^2 + bt/3\right)}\right),$$

and the same bound holds for the left tail.

Proof. A proof of a slightly more general inequality is the object of Exercise 5. □

In contrast with the Multiplicative Chernoff, the Bernstein bound is valid for any bounded random variables (not just Bernoulli). As opposed to Hoeffding, the dominating term scales with the right order $\sqrt{n\,\mathrm{var}(Z)}$. As a consequence, involving a 2-terms upper bound, the Bernstein might appear not so convenient at first glance, it remains however a safe and general approach

as the rate of convergence is the good one. Let $s = \log(1/\delta)$. We look for $t > 0$ such that $\dfrac{t^2}{2(n\sigma^2 + bt/3)} = s$, that is,

$$t^2 - \frac{2bs}{3}\, t - 2n\sigma^2 s = 0.$$

This quadratic has a unique positive root,

$$t_\delta = \frac{bs}{3} + \sqrt{\frac{b^2 s^2}{9} + 2n\sigma^2 s}.$$

Since $t_\delta > 0$ solves the equation, the tail bound at $t = t_\delta$ equals $e^{-s} = \delta$, so $\mathbb{P}(S > t_\delta) \leq \delta$. Using $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ for $x, y \geq 0$,

$$t_\delta \leq \frac{2b}{3} \log(1/\delta) + \sqrt{2n\sigma^2 \log(1/\delta)}.$$

Hence, with probability at least $1 - \delta$,

$$\sum_{i=1}^{n} (Z_i - \mathbb{E}Z) \leq \sqrt{2n\sigma^2 \log(1/\delta)} + \frac{2b}{3} \log(1/\delta).$$

For the two-sided version, apply the bound to Z_i and $-Z_i$ and use the union bound, which replaces $\log(1/\delta)$ by $\log(2/\delta)$: with probability at least $1 - \delta$,

$$\Big| \sum_{i=1}^{n} (Z_i - \mathbb{E}Z) \Big| \leq \sqrt{2n\sigma^2 \log(2/\delta)} + \frac{2b}{3} \log(2/\delta).$$

Finally, it is useful to consider the max of sub-Gaussian random variables (instead of the sum). This is the topic of the following result where the independence of the collection Z_k is not needed any more, in contrast to previous results dealing with sums.

Lemma 7. *Let $(Z_i)_{i=1,\dots,M}$ be a collection of sub-Gaussian random variables with common sub-Gaussian factor $v > 0$. It holds that*

$$\mathbb{E}[\max_{k=1,\dots,M} Z_k] \leq \sqrt{2v \log(M)}$$

and with probability $1 - \delta$,

$$\max_{k=1,\dots,M} Z_k \leq \sqrt{2v \log(M/\delta)}.$$

Proof. For any $\lambda > 0$, it holds

$$\max_{k=1,\dots,M} Z_k = \lambda^{-1} \log(\exp(\lambda \max_{k=1,\dots,M} Z_k)) \leq \lambda^{-1} \log(\sum_{k=1}^{M} \exp(\lambda Z_k)).$$

Jensen's inequality and the sub-Gaussian property imply that

$$\mathbb{E}[\max_{k=1,\dots,M} Z_k] \leq \lambda^{-1} \log(\sum_{k=1}^{M} \mathbb{E}[\exp(\lambda Z_k)]) \leq \lambda^{-1} \log(M \exp(\lambda^2 v/2)) = \lambda^{-1} \log(M) + \lambda v/2.$$

Optimizing in λ gives $\lambda = \sqrt{2\log(M)/v}$ and we obtain the first statement. The upper bound in probability is obtained from the union bound. □

2.3 Application: Pointwise estimation with k-nearest neighbors

2.3.1 Background

Let $(X, Y), (X_1, Y_1), \ldots, (X_n, Y_n)$ be a collection of independent and identically distributed random variables such that $X \in \mathbb{R}^d$ is the covariate and $Y \in \mathbb{R}$. We suppose that $\mathbb{E}[|Y|]$ exists. Let $h(X) = \mathbb{E}[Y|X]$. Let $x \in \mathbb{R}^d$. We further suppose that h (has a version that) is Lipschitz at x. This is stated below.

(H) $((X, Y), (X_1, Y_1), \ldots, (X_n, Y_n))$ is a collection of independent and identically distributed random elements with common distribution P on $\mathbb{R}^d \times \mathbb{R}$ and $E[|Y|] < +\infty$. The conditional expectation $h(X) = \mathbb{E}[Y|X] \in L_1(P)$ has a version satisfying $|h(y) - h(x)| \leq L\|y - x\|$ for all $y \in B(x, \tau_0)$, where $\tau_0 > 0$.

Let $k \in \{1, \ldots, n\}$, the k-nearest neighbor (k-NN for short) regression estimate of $h(x) = \mathbb{E}[Y|X = x]$ at point $x \in \mathbb{R}^d$ is defined as

$$\hat{h}(x) = k^{-1} \sum_{i \in N_k(x)} Y_i,$$

where $N_k(x)$ is the index set of the k-nearest neighbor to point x among $X_1, \ldots, X_n$ (if there are some ties use the lexicographic order). The k-NN algorithm was initially introduced in Fix and Hodges (1951); Royall (1966); Cover and Hart (1967) and has since been the subject of many studies in the statistical and machine learning literature. For a detailed exposition of the main mathematical results on k-NN estimation (including density, regression and classification), we refer to the textbooks Györfi et al. (2006) and Biau and Devroye (2015).

Our goal here is to obtain a deviation inequality for the k-NN error $|\hat{h}(x) - h(x)|$. The analysis is only based on the sub-Gaussian bound and the Chernoff multiplicative bound from the sections before. The approach taken relies on the following decomposition:

$$\hat{h}(x) - h(x) = k^{-1} \sum_{i \in N_k(x)} (Y_i - h(X_i)) + k^{-1} \sum_{i \in N_k(x)} (h(X_i) - h(x)).$$

While the second term, called the bias term, will be successfully treated with the help of the multiplicative Chernoff bound, the first term, called the stochastic term, will require the use of the sub-Gaussian bound.

2.3.2 The bias term

We suppose that the covariates distribution admits a density putting some positive mass on a ball around point x.

(X) The distribution of X admits a density f on $\mathbb{R}^d$. There is $\tau_0 > 0$ such that $b_0 := \inf_{y \in B(x, \tau_0)} f(y) > 0$.

The k-NN bandwidth at x is denoted by $\hat{\tau}_{k,x}$ and defined as the smallest radius $\tau \geq 0$ such that the ball $B(x, \tau)$ contains at least k points from the collection $\{X_1, \ldots, X_n\}$. That is,

$$\hat{\tau}_{k,x} := \inf\{\tau \geq 0 : \sum_{i=1}^{n} \mathbb{1}_{B(x,\tau)}(X_i) \geq k\}.$$

The k-NN bandwidth at x is $\hat{\tau}_{n,k,x} := \inf\{\tau \geq 0 : \sum_{i=1}^n \mathbf{1}_{B(x,\tau)}(X_i) \geq k\}$, where $B(x,\tau)$ denotes the closed ball and $1 \leq k \leq n$. Writing $D_{(1)} \leq \cdots \leq D_{(n)}$ for the ordered distances $\|X_i - x\|$, the set is $[D_{(k)}, \infty)$, so the infimum is attained and $\hat{\tau}_{n,k,x} = D_{(k)}$. We have the following result showing that the size of $\hat{\tau}_{k,x}$ is at most of order $(k/n)^{1/d}$.

Lemma 8. *Suppose that (H) and (X) hold true. For all $n \geq 1$, $\delta \in (0,1)$ and $1 \leq k \leq n$ such that $4\log(1/\delta) \leq k \leq n\tau_0^d b_0 V_d/2$, it holds, with probability at least $1-\delta$:*

$$\hat{\tau}_{k,x} \leq \left(\frac{2k}{nb_0V_d}\right)^{1/d}.$$

Proof. Define $\overline{\tau}_k = (2k/(nb_0V_d))^{1/d}$. The assumption $k \leq n\tau_0^d b_0 V_d/2$ gives $\overline{\tau}_k \leq \tau_0$, hence $\mu := P(B(x,\overline{\tau}_k)) \geq b_0 V_d \overline{\tau}_k^d = 2k/n$. By definition, $\hat{\tau}_k \leq \overline{\tau}_k$ as soon as $\sum_{i=1}^n \mathbb{1}_{B(x,\overline{\tau}_k)}(X_i) \geq k$. Therefore, we only need to show that the previous happens with probability $1-\delta$. By multiplicative Chernoff, Theorem 1, it holds with probability $1-\delta$,

$$\sum_{i=1}^n \mathbb{1}_{B(x,\overline{\tau}_k)}(X_i) \geq n\mu - \sqrt{2n\mu\log(1/\delta)}$$

But we have shown that $n\mu \geq 2k$. The function $y \mapsto y - \sqrt{2yl}$ is increasing whenever $y \geq l/2$. Hence if $2k \geq (1/2)\log(1/\delta)$ (which is implied by $k \geq 4\log(1/\delta)$), we obtain that

$$\sum_{i=1}^n \mathbb{1}_{B(x,\overline{\tau}_k)}(X_i) \geq n\mu - \sqrt{2n\mu\log(1/\delta)} \geq 2k - \sqrt{4k\log(1/\delta)}$$

The above is larger than k whenever $k \geq \sqrt{4k\log(1/\delta)}$ or $k \geq 4\log(1/\delta)$. □

As a consequence of the previous Lemma, we have that, under the stipulated assumption, it holds, with probability $1-\delta$,

$$\left| k^{-1} \sum_{i \in N_k(x)} (h(X_i) - h(x)) \right| \leq L\left(\frac{2k}{nb_0V_d}\right)^{1/d}.$$

This bound is often referred to as the bias-bound in the analysis of k-NN regression. We note that the smaller k the better the bound.

2.3.3 The stochastic term

We now consider the stochastic term. The following assumption will allow us to use a sub-Gaussian inequality.

(E) For each $i = 1, \ldots, n$, $\epsilon_i = Y_i - h(X_i)$ is sub-Gaussian with factor σ^2 conditionally on X_i, i.e., $\mathbb{E}[\exp(\lambda\epsilon_i)|X_i] \leq \exp(\lambda^2\sigma^2/2)$ for all $\lambda \in \mathbb{R}$ and $\mathbb{E}[\epsilon_i|X_i] = 0$.

The above assumption implies that ϵ_i is sub-Gaussian with factor σ^2. Moreover, note that this assumption is valid, for instance, when ϵ_i is independent from X_i and is sub-Gaussian with factor σ^2. Finally, one could also allow σ^2 to depend on X_i (at the price of some slight changes in the next statement).

Theorem 9. *Suppose that (H) and (E) hold true. For all $n \geq 1$, $\delta \in (0,1)$ and $1 \leq k \leq n$, we have, with probability $1-\delta$,*

$$\left|\hat{h}(x) - h(x)\right| \leq \sqrt{\frac{2\sigma^2}{k}\log(2/\delta)}.$$

Proof. We have that $(\epsilon_i)_{i=1,\dots,n}$ is an independent collection of random variable conditionally on $(X_i)_{i=1,\dots,n}$. Consider the conditional probability $\mathbb{P}_\epsilon$ given $(X_i)_{i=1,\dots,n}$. It follows that

$$\mathbb{E}_\epsilon \exp\left(\lambda \sum_{i \in N_k(x)} \epsilon_i\right) = \prod_{i \in N_k(x)} \mathbb{E}_\epsilon \exp(\lambda \epsilon_i) = \prod_{i \in N_k(x)} \mathbb{E}[\exp(\lambda \epsilon_i)|X_i].$$

Using (E) gives

$$\mathbb{E}_\epsilon \exp\left(\lambda \sum_{i \in N_k(x)} \epsilon_i\right) \leq \prod_{i \in N_k(x)} \exp(\lambda^2\sigma^2/2) \leq \exp(\lambda^2 k\sigma^2/2).$$

We have that $\sum_{i \in N_k(x)} \epsilon_i$ is sub-Gaussian with factor $k\sigma^2$, with respect to $\mathbb{P}_\epsilon$. From Theorem 4, we obtain

$$\mathbb{P}_\epsilon\left(\left|\sum_{i \in N_k(x)} \epsilon_i\right| > t\right) \leq 2\exp(-t^2/(2k\sigma^2)).$$

In other words,

$$\mathbb{P}_\epsilon\left(\left|\sum_{i \in N_k(x)} \epsilon_i\right| > \sqrt{2k\sigma^2\log(2/\delta)}\right) \leq \delta.$$

Taking the expectation gives

$$\mathbb{P}\left(\left|\sum_{i \in N_k(x)} \epsilon_i\right| > \sqrt{2k\sigma^2\log(2/\delta)}\right) \leq \delta.$$

□

2.3.4 Final bound

Now we can state the main result concerning the pointwise analysis of k-NN regression. The next result is a straightforward consequence of the previous analysis on the bias and stochastic terms, using the union bound to conclude.

Theorem 10. *Suppose that (H), (X) and (E) hold true and that h is L-Lipschitz. For all $n \geq 1$, $\delta \in (0,1/2)$ and $1 \leq k \leq n$ such that $4\log(1/\delta) \leq k \leq n\tau_0^d b_0 V_d/2$, we have, with probability $1-2\delta$,*

$$\left|\hat{h}(x) - h(x)\right| \leq \sqrt{\frac{2\sigma^2}{k}\log(2/\delta)} + L\left(\frac{2k}{nb_0V_d}\right)^{1/d}.$$

2.4 Exercises

Exercise 1 (improving the multiplicative Chernoff bound)**.** *Let $(Z_i)_{i=1,\dots,n}$ be i.i.d. $\mathcal{B}(\mu)$ with $\mu \in (0,1)$, and let $\delta \in (0,1)$.*

1. *Show that, with probability at least $1-\delta$,*
$$\sum_{i=1}^{n}(Z_i - \mu) \le \sqrt{2n(1-\mu)\log(1/\delta)}.$$
2. *Deduce that when $\mu \ge 2/5$, this bound is smaller than the upper bound*
$$3\log(1/\delta) \vee \sqrt{3n\mu\log(1/\delta)}$$
obtained from Theorem 1, whatever the value of n and δ.
3. *Show that, when $\mu < 2/5$ and $n\mu \ge 3\log(1/\delta)$, the comparison is reversed: the bound of Theorem 1 is smaller.*

Exercise 2. *Let Z be sub-Gaussian with factor $v > 0$, i.e. $\mathbb{E}[Z] = 0$ and $\mathbb{E}[e^{\lambda Z}] \le e^{\lambda^2 v/2}$ for all $\lambda \in \mathbb{R}$. Show that $\mathbb{E}[Z^2] = \mathrm{var}(Z) \le v$ (*Hint. *Use that $e^x - 1 - x \ge 0$ for all $x \in \mathbb{R}$, and apply Fatou's lemma to $(e^{\lambda Z} - 1 - \lambda Z)/\lambda^2$ as $\lambda \to 0$).*

Exercise 3 (examples of sub-Gaussian random variables)**.** *Show that $Z \sim \mathcal{N}(0,1)$ is subGaussian with sub-Gaussian factor $v = 1$. Show that $Z \sim \mathcal{N}(0,v)$ is sub-Gaussian with sub-Gaussian factor v. Show that whenever $E[Z] = 0$ and $a \le Z \le b$ almost surely for $a \le 0 \le b$, then Z is sub-Gaussian with sub-Gaussian factor $(b-a)^2/4$. One may introduce $\psi(\lambda) = \log(\mathbb{E}[\exp(\lambda Z)])$ and use that the second derivative is the variance of a measure dominated by the one of Z. (hint: note that $\mathrm{var}(Z) \le (b-a)^2/4$). Show that whenever $a \le Z \le b$ then $Z - \mathbb{E}[Z]$ is sub-Gaussian with sub-Gaussian factor $(b-a)^2/4$.*

Exercise 4 (multiplicative Chernoff and sub-Gaussian)**.** *Let $(Z_i)_{1\le i\le n}$ be a sequence of independent Bernoulli random variables with mean μ. Compare the two upper bounds obtained (with probability $1-\delta$) from multiplicative Chernoff and sub-Gaussian. Show that multiplicative Chernoff is better whenever $\mu \le 1/6$.*

Exercise 5 (Bernstein's proof)**.** *Let $Z_1, \dots, Z_n$ be i.i.d. with $\mathrm{var}(Z_1) = \sigma^2$ and $|Z_1 - \mathbb{E}Z_1| \le b$ almost surely. Set $Y_i = Z_i - \mathbb{E}Z_i$, $S = \sum_{i=1}^n Y_i$ and $v = n\sigma^2$.*

1. *Show that $\mathbb{E}[|Y_1|^k] \le \sigma^2 b^{k-2}$ for all $k \ge 2$, and that $k! \ge 2 \cdot 3^{k-2}$.*
2. *Show that for all $0 < \lambda < 3/b$,*
$$\mathbb{E}\, e^{\lambda Y_1} \le \exp\Big(\frac{\lambda^2\sigma^2}{2(1-\lambda b/3)}\Big), \qquad \textit{hence} \qquad \mathbb{E}\, e^{\lambda S} \le \exp\Big(\frac{\lambda^2 v}{2(1-\lambda b/3)}\Big).$$
Hint. *Expand $e^{\lambda Y_1}$ in a power series, use $\mathbb{E}Y_1 = 0$, step 1, and $1 + u \le e^u$.*
3. *Let $s > 0$ and $c = b/3$. Show that*
$$\mathbb{P}\big(S > \sqrt{2vs} + cs\big) \le e^{-s}.$$
Hint. *Apply Chernoff's method with $\lambda = a/(1+ca)$, where $a = \sqrt{2s/v}$ (note $\lambda < 1/c$), and check that $\lambda t - \frac{\lambda^2 v}{2(1-c\lambda)} = s$ for $t = \sqrt{2vs} + cs$.*

4. Deduce that, for $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\Big|\sum_{i=1}^{n}(Z_i - \mathbb{E}Z_1)\Big| \leq \sqrt{2n\sigma^2 \log(2/\delta)} + \frac{b}{3}\log(2/\delta).$$

Exercise 6 (from tail to expectation)**.** *Show that $E[|X|] = \int_0^\infty P(|X| > t)dt$. Let $(Z_i)_{i\geq 1}$ be a collection of independent random variables such that each $Z_i - \mathbb{E}[Z_i]$ is sub-Gaussian with factor v_i. Show that*

$$\mathbb{E}|\sum_{i=1}^{n}(Z_i - \mathbb{E}Z_i)| \leq \sqrt{2\pi V},$$

where $V = \sum_{i=1}^n v_i$. Using that $\exp(|x|) \leq \exp(x) + \exp(-x)$ and the definition of sub-Gaussian random variables, show that the previous bound can be improved to

$$\mathbb{E}|\sum_{i=1}^{n}(Z_i - \mathbb{E}Z_i)| \leq \sqrt{2V\log(2)}.$$

Exercise 7. *Show that when $(Z_i)_{i=1,\dots,M}$ is a collection of sub-Gaussian random variables with common sub-Gaussian factor $v > 0$, we have*

$$\mathbb{E}[\max_{k=1,\dots,M}|Z_k|] \leq \sqrt{2v\log(2M)},$$

and with probability $1-\delta$,

$$\max_{k=1,\dots,M}|Z_k| \leq \sqrt{2v\log(2M/\delta)}.$$

Exercise 8 (KDE)**.** *Let $x \in \mathbb{R}^d$. Let $\hat{f}_n(x)$ be the KDE estimate of $f(x)$, a L_X-Lipschitz density, based on the uniform Kernel, i.e., $K(x) \propto \mathbb{1}_{\{B(0,1)\}}(x)$. Let $V_d = \int_{B(0,1)} du$. Define $f_h(x) = \mathbb{E}[\hat{f}_n(x)]$. Suppose that h is sufficiently small, i.e., $f_h(x) \in (f(x)/2, 2f(x))$. Show that with probability $1-\delta$,*

$$|\hat{f}_n(x) - f(x)| \leq hL_X + \sqrt{\frac{6f(x)}{nh^dV_d}\log(2/\delta)}.$$

Conclude arguing that using Hoeffding inequality, one would obtain a poor scaling for the variance term in nh^{2d}.

Exercise 9 (finite ERM)**.** *Let $2, (Z_i)_{i\geq 1}$ be an independent and identically distributed sequence of random variables with common distribution P on $\mathcal{Z}$. Let $\mathcal{H} = \{h_1, \dots, h_M\}$ be a set of prediction rules that are evaluated through the loss function $\ell : \mathcal{Z}\times\mathcal{H} \to [0,1]$. Define $\hat{h}_n \in \arg\min_{h\in\mathcal{H}} \hat{R}_n(h)$ where $\hat{R}_n(h) = n^{-1}\sum_{i=1}^n \ell(Z_i, h)$ as well as $h^* \in \arg\min_{h\in\mathcal{H}} R(h)$ where $R(h) = \mathbb{E}[\ell(Z,h)]$. Then for all $\delta \in (0,1)$, it holds with probability $1-\delta$,*

$$R(\hat{h}_n) \leq R(h^*) + \sqrt{\frac{2}{n}\log((M+1)/\delta)}$$

One can start noting that

$$R(\hat{h}_n) - R(h^*) = R(\hat{h}_n) - \hat{R}_n(\hat{h}_n) + \hat{R}_n(\hat{h}_n) - \hat{R}_n(h^*) + \hat{R}_n(h^*) - R(h^*)$$

where $\hat{R}_n(\hat{h}_n) - \hat{R}_n(h^*) \leq 0$ by definition.

Exercise 10 (OLS)**.** *In the fixed-design linear model with sub-Gaussian noise, i.e., $Y_i = x_i^T\theta + \epsilon_i$, where x_i are real numbers and ϵ_i are sub-Gaussian random variables with factor σ^2, show that*

$$R(\hat{\boldsymbol{\theta}}_n) \leq R(\boldsymbol{\theta}^*) + \frac{2(p+1)\sigma^2}{n}\log(2(p+1)/\delta).$$

3 Empirical processes, symmetrization and Rademacher complexity

3.1 Background

We shall rely on the following assumption.

(Ziid) $(Z, Z_1, \ldots, Z_n)$ is a collection of random variables independent and identically distributed with common distribution P on $(S, \mathcal{S})$.

We consider a class $\mathcal{G}$ of measurable real-valued functions g defined on S. The aim is to obtain upper bounds, valid under certain probability, on the random variable

$$Z_{\mathcal{G}} = \sup_{g \in \mathcal{G}} |Z_g| ,$$

where

$$Z_g = \sum_{i=1}^{n} (g(Z_i) - \mathbb{E}[g(Z)]).$$

Similar bounds have been establish in previous chapter for a single function g or finitely many as in Lemma 7. From now on, the difficulty will be to study the case were the class $\mathcal{G}$ includes infinitely many elements.

The class $\mathcal{G}$ is supposed to have countably many elements $g \in \mathcal{G}$ to avoid any measurability problem related to the suprema operation. Indeed, a countable supremum of measurable functions is always measurable. In many practical cases, the measurability can be obtained using some other particular properties related to the problem of interest. For instance, the class $\mathcal{G}$ can be uncountable if one has the separability of the underlying process, see for instance Van Handel (2014), which basically ensures that there is $\mathcal{G}_0$ countable so that, almost surely,

$$\sup_{g \in \mathcal{G}} |Z_g| = \sup_{g \in \mathcal{G}_0} |Z_g| .$$

As suggested by Van Der Vaart and Wellner (2007), Example 2.3.4, point-wise measurability, i.e., for every g, there is $g_n \in \mathcal{G}_0$ such that for every x, $\lim g_n(x) = g(x)$, is enough to imply separability. Another way around this problem is to work with the outer expectation as explained in Van Der Vaart and Wellner (2007).

Important quantities in the sequel will be the uniform bound on the class $\mathcal{G}$ and the greatest variance value among the class $\mathcal{G}$. They are introduced in the following assumption.

(B) There is $U > 0$ and $\sigma > 0$ such that for all $g \in \mathcal{G}$,

$$\sup_{x \in S} |g(x)| \leq U, \qquad \operatorname{var}(g(Z)) \leq \sigma^2.$$

3.2 Symmetrization

An important property in what follows is that $\sup_{g \in \mathcal{G}} |Z_g|$ behaves similarly to a symmetrized version defined as

$$\sup_{g \in \mathcal{G}} |Z_g - Z'_g|$$

where

$$Z'_g = \sum_{i=1}^{n} (g(Z'_i) - P(g))$$

is based on another independent collection of random variables $(Z'_i)_{1\leq i\leq n}$, called the ghost collection, having the same distribution as the original $(Z_i)_{1\leq i\leq n}$. Symmetrization is also refereed to as randomization because $Z_g - Z'_g$ has the same distribution as the randomized quantity $|R_g - R'_g|$ with

$$R_g := \sum_{i=1}^{n} \eta_i g(Z_i) \text{ and } R'_g := \sum_{i=1}^{n} \eta_i g(Z'_i).$$

The collection $(\eta_1, \ldots, \eta_n)$ is a collection of independent and identically distributed variables such that $\mathbb{P}(\eta_1 = 1) = 1/2 = \mathbb{P}(\eta_1 = -1)$. They are called Rademacher variables. The next statement is dealing with the expectation.

Lemma 11 (symmetrization of expectation)**.** *Under (Ziid), suppose that $(Z'_i)_{1\leq i\leq n}$ is another collection independent from $(Z_i)_{1\leq i\leq 1}$ with the same distribution. We have*

$$\mathbb{E}[Z_{\mathcal{G}}] = \mathbb{E}[\sup_{g\in\mathcal{G}} |Z_g - Z'_g|] \leq 2\mathbb{E}[Z_{\mathcal{G}}].$$

Moreover

$$\mathbb{E}[\sup_{g\in\mathcal{G}} |Z_g - Z'_g|] = \mathbb{E}[\sup_{g\in\mathcal{G}} |R_g - R'_g|] \leq 2\mathbb{E}[\sup_{g\in\mathcal{G}} |R_g|]$$

Proof. In what follows, E_W denote the expectation with respect to W. Write

$$\begin{aligned}\mathbb{E}[Z_{\mathcal{G}}] &= \mathbb{E}_Z[\sup_{g\in\mathcal{G}} \left| E_{Z'}[\sum_{i=1}^{n} (g(Z_i) - g(Z'_i))] \right|] \\ &\leq \mathbb{E}_Z \mathbb{E}_{Z'}[\sup_{g\in\mathcal{G}} \left| \sum_{i=1}^{n} (g(Z_i) - g(Z'_i)) \right|]\end{aligned}$$

The first inequality follows from the remark that if W is symmetric, then ηW has the same distribution as W. The second inequality follows from the triangle inequality.

□

Note that other symmetrization inequalities exist. For instance, a more general symmetrization result, valid for convex and increasing functions (instead of the above absolute value function) is explored in Boucheron et al. (2013). Let us give the following symmetrization principle (see (Bousquet et al., 2004, lemma 2) and (Devroye et al., 2013, proof of Theorem 12.4)), based on deviation probability instead of expected value.

Lemma 12 (symmetrization of probability)**.** *Under (Ziid), suppose that $(Z'_i)_{1\leq i\leq n}$ is another collection independent from $(Z_i)_{1\leq i\leq 1}$ with the same distribution. If $a \leq g \leq b$ and $2n(b-a)^2 \leq \epsilon^2$, we have*

$$\mathbb{P}[Z_{\mathcal{G}} > \epsilon] \leq 2\mathbb{P}\left(\sup_{g\in\mathcal{G}} \left|Z_g - Z'_g\right| > \epsilon/2\right) \leq 4\mathbb{P}[Z_{\mathcal{G}} > \epsilon/4].$$

Proof. Start the proof remarking that $A := \{Z_{\mathcal{G}} > \epsilon\}$ if and only if there is $g^* \in \mathcal{G}$ such as $Z_{g^*} > \epsilon$. Under A, let us define g^* such that $Z_{g^*} > \epsilon$. We have $|Z_{g^*}| - |Z'_{g^*}| \leq |Z_{g^*} - Z'_{g^*}| \leq \sup_g |Z_g - Z'_g|$. But $|Z_{g^*}| > \epsilon$ and $|Z'_{g^*}| \leq \epsilon/2$ implies that $|Z_{g^*}| - |Z'_{g^*}| \geq \epsilon/2$ and therefore $\sup_g |Z_g - Z'_g| > \epsilon/2$. Hence

$$\mathbb{P}(A, |Z'_{g^*}| \leq \epsilon/2) \leq \mathbb{P}(\sup_g |Z_g - Z'_g| > \epsilon/2).$$

By Chebyshev inequality, we have $\mathbb{P}'(|Z'_{g^*}| > \epsilon/2) \leq (4n/\epsilon^2) \operatorname{var}_{Z'}(g^*(Z')) \leq n(b-a)^2/\epsilon^2 \leq 1/2$. It follows that

$$\mathbb{P}(A, |Z'_{g^*}| \leq \epsilon/2) \geq (1/2)\mathbb{P}(A).$$

□

The next lemma is also about symmetrization. The proof relies almost solely on the convexity of Ψ making it different from the previous one.

Lemma 13 (symmetrization with convex function)**.** *Under (Ziid), suppose that $(Z'_i)_{1\leq i\leq n}$ is another collection independent from $(Z_i)_{1\leq i\leq 1}$ with the same distribution. Let $\Psi : \mathbb{R} \to \mathbb{R}_{\geq 0}$ be increasing and convex. Let $Z_g = \sum_{i=1}^n \{g(Z_i) - Eg(Z)\}$ and $Z'_g = \sum_{i=1}^n \{g(Z'_i) - Eg(Z)\}$. We have*

$$\mathbb{E}_Z\left[\Psi(\sup_{g\in\mathcal{G}} |Z_g|)\right] \leq \mathbb{E}_Z\mathbb{E}_{Z'}\left[\Psi(\sup_{g\in\mathcal{G}} \left|Z_g - Z'_g\right|)\right] \leq \mathbb{E}_Z\left[\Psi(2\sup_{g\in\mathcal{G}} |Z_g|)\right]$$

and

$$\mathbb{E}_Z\left[\Psi(\sup_{g\in\mathcal{G}} Z_g)\right] \leq \mathbb{E}_Z\mathbb{E}_{Z'}\left[\Psi(\sup_{g\in\mathcal{G}} Z_g - Z'_g)\right] \leq \mathbb{E}_Z\left[\Psi(2\sup_{g\in\mathcal{G}} |Z_g|)\right].$$

Moreover

$$\mathbb{E}_Z\left[\Psi(\sup_{g\in\mathcal{G}} |Z_g|)\right] \leq \mathbb{E}_Z\left[\Psi(2\sup_{g\in\mathcal{G}} |R_g|)\right]$$

and

$$\mathbb{E}_Z\left[\Psi(\sup_{g\in\mathcal{G}} Z_g)\right] \leq \mathbb{E}_Z\left[\Psi(2\sup_{g\in\mathcal{G}} R_g)\right].$$

Proof. See Exercise 12. □

Lemma 14 (symmetrization of probability with relative deviation)**.** *Under (Ziid), suppose that $(Z'_i)_{1\leq i\leq n}$ is another collection independent from $(Z_i)_{1\leq i\leq 1}$ with the same distribution. If $n\epsilon^2 > 2$, we have*

$$\mathbb{P}\left(\exists g \in \mathcal{G} \,:\, -Z_g > \epsilon\sqrt{nP(g)}\right) \leq$$
$$4\mathbb{P}\left(\exists g \in \mathcal{G} \,:\, Z_g - Z'_g > \epsilon\sqrt{(1/2)(Z_g + Z_g)}\right).$$

Proof. Let $A := \exists g, \sum_{i=1}^n (P(g) - g(Z_i)) > \epsilon\sqrt{nP(g)}$. Under A, define g^* such that $\sum_{i=1}^n P(g^*) - g^*(Z_i) > \epsilon\sqrt{nP(g^*)}$. Let $B := \sum_{i=1}^n g^*(Z_i') - P(g^*) > 0$ and define

$$F := \frac{\sum_{i=1}^n (g^*(Z_i') - g^*(Z_i))}{\sqrt{(1/2)\sum_{i=1}^n (g^*(Z_i)^2 + g^*(Z_i')^2)}}.$$

On $A \cap B$, we have, using that $x \mapsto (x-a)/\sqrt{x+a}$ is increasing,

$$F > \frac{\sum_{i=1}^n (P(g^*) - g^*(Z_i))}{\sqrt{(1/2)\sum_{i=1}^n (g^*(Z_i)^2 + P(g^*))}} > \frac{\sum_{i=1}^n (P(g^*) - g^*(Z_i))}{\sqrt{nP(g^*)}} > \epsilon.$$

Note that since $P_n(g^*) \geq 0$, we have $P(g^*) > \epsilon^2$ and therefore $P(g^*) > 2/n$. Since $P'(B) > 1/4$. It follows $1_A \leq 1_A 4P'(B) = 4P'(A,B) \leq 4P'(F > \epsilon)$. Conclude by taking the expectation. □

3.3 Rademacher complexity

The so-called Rademacher complexity is defined as follows. Let $(\eta_i)_{1 \leq i \leq n}$ denote a collection of independent Rademacher variables, that is, $\mathbb{P}(\eta_i = +1) = \mathbb{P}(\eta_i = -1) = 1/2$. The Rademacher complexities (the one-sided and the other) are given by

$$\mathbb{E}_\eta[\sup_{g \in \mathcal{G}} \sum_{i=1}^n \eta_i g(Z_i)] \quad \text{and} \quad \mathbb{E}_\eta[\sup_{g \in \mathcal{G}} \left| \sum_{i=1}^n \eta_i g(Z_i) \right|].$$

where $\mathbb{E}_\eta$ is over the Rademacher variables. In our context, the Rademacher complexity is useful because of the following inequality:

$$\mathbb{E}[Z_{\mathcal{G}}] \leq 2\mathbb{E}_\eta[\sup_{g \in \mathcal{G}} \left| \sum_{i=1}^n \eta_i g(Z_i) \right|],$$

which follows from the symmetrization lemma. Another Gaussian (instead of Bernoulli) randomization result is given in Van de Geer et al. (2016). Depending on the situation, we might use one of the above preferably to the other. Here are some properties on Rademacher complexity.

Lemma 15. *Let* $\mathrm{Rad}_n(\mathcal{F}) = \mathbb{E}_\eta[\sup_{g \in \mathcal{F}} |\sum_{i=1}^n \eta_i g(Z_i)|]$. *Under (Ziid), it holds*

1. *If* $\mathcal{F} \subset \mathcal{G}$ *then* $\mathrm{Rad}_n(\mathcal{F}) \leq \mathrm{Rad}_n(\mathcal{G})$.
2. $\mathrm{Rad}_n(c\mathcal{G}) = |c|\mathrm{Rad}_n(\mathcal{G})$.
3. $\mathrm{Rad}_n(\mathcal{G} + g_0) \leq \mathrm{Rad}_n(\mathcal{G}) + \|g_0\|_\infty n^{1/2}$.
4. *Let* φ *be* L*-Lipschitz, i.e., for all* x, y, $|\varphi(x) - \varphi(y)| \leq L|x - y|$ *and* $\varphi(0) = 0$, *then* $\mathrm{Rad}_n(\varphi(\mathcal{G})) \leq 2L\mathrm{Rad}_n(\mathcal{G})$. *For the one-sided Rademacher (without absolute value), the inequality is valid with* L *instead of* $2L$.

The last inequality is referred to as the contraction principle, due to (Ledoux and Talagrand, 1991). A proof is given in Boucheron et al. (2013), Theorem 11.6. It will be useful to obtain a bound on the empirical variance $\sum_{i=1}^n g(X_i)^2$.

3.4 From expectation to deviation probability: Bousquet's concentration inequality

Bousquet's inequality plays a crucial role in statistical learning theory and especially when dealing with functional estimators or empirical risk minimization. It allows for instance, to obtain sharp deviation bound for local regression estimators or training classifiers based on empirical risk minimization. The next statement is Theorem 2.3 from Bousquet (2002) but other versions may be found in Talagrand (1996); Massart (2000); Rio (2002).

Theorem 16. *Under (Ziid) and (B), suppose that* $|g| \leq 1$ *and* $P(g) = 0$. *We have for all* $\delta \in (0,1)$,

$$P\left(Z_{\mathcal{G}} - \mathbb{E}Z_{\mathcal{G}} \geq \sqrt{2s\log(1/\delta)} + \log(1/\delta)/3\right) \leq \delta.$$

where $s = 2\mathbb{E}Z_{\mathcal{G}} + n\sigma^2$. *The same holds for* $Z_{\mathcal{G}}$ *defined without the absolute value.*

Next statement is just a weaker version of the previous one allowing to consider functions bounded by U (instead of 1) and that have expected value different from 0. The terms have been re-arranged to get rid of s and only have $\mathbb{E}Z_{\mathcal{G}}$, σ^2 and U.

Theorem 17. *Under (Ziid) and (B), we have with probability* $1-\delta$,

$$Z_{\mathcal{G}} \leq 2\mathbb{E}[Z_{\mathcal{G}}] + \sqrt{2n\sigma^2\log(1/\delta)} + 8U\log(1/\delta)/3.$$

The same holds for $Z_{\mathcal{G}}$ *defined without the absolute value.*

Proof. Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, with probability $1-\delta$, it holds

$$Z_{\mathcal{G}} - \mathbb{E}Z_{\mathcal{G}} \leq 2\sqrt{\mathbb{E}Z_{\mathcal{G}}\log(1/\delta)} + \sqrt{2n\sigma^2\log(1/\delta)} + \log(1/\delta)/3,$$

and invoking that $2ab \leq a^2 + b^2$, we obtain

$$Z_{\mathcal{G}} - \mathbb{E}Z_{\mathcal{G}} \leq \mathbb{E}Z_{\mathcal{G}} + \sqrt{2n\sigma^2\log(1/\delta)} + 4\log(1/\delta)/3.$$

As a consequence we have with probability $1-\delta$,

$$Z_{\mathcal{G}} \leq 2\mathbb{E}Z_{\mathcal{G}} + \sqrt{2n\sigma^2\log(1/\delta)} + 4\log(1/\delta)/3.$$

We now apply the previous inequality to $(g - P(g))/2U$ and we get the stated result.

□

The upper bound in the previous theorem consists of three terms. The third term is asymptotically negligible, while the second exhibits the standard $\sqrt{n}$ convergence rate and variance dependence seen in multiplicative Chernoff (Theorem 1) and some other Bernstein-type inequalities. To obtain a meaningful deviation bound for $Z_{\mathcal{G}}$, it therefore remains to analyse the first term, $\mathbb{E}[Z_{\mathcal{G}}]$.

Bousquet's inequality underlines an important way to analyse statistical learning problems. It suggests to rely on the following decomposition

$$Z_{\mathcal{G}} = (Z_{\mathcal{G}} - \mathbb{E}Z_{\mathcal{G}}) + \mathbb{E}Z_{\mathcal{G}}.$$

The first part, $Z_{\mathcal{G}} - \mathbb{E}Z_{\mathcal{G}}$, is examined with the help of Bousquet's inequality. The second part, $\mathbb{E}Z_{\mathcal{G}}$, can be studied using a symmetrization arguments, as in 11, coupled with the Vapnik inequalities given in the the next section or relying on Dudley's entropy integral as in Theorem 35 given in Section 5.

We note that when relying on the symmetrization inequalities valid for the deviation probability as Lemma 12 and Lemma 14, the Bousquet's inequality may not be needed to obtain estimate on the deviation probability.

3.5 Exercises

Exercise 11. *Show that the results on symmetrization (of the expectation and the probability) are also valid without absolute values, i.e.,*

$$\mathbb{E}[\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}(g(Z_i)-Pg)] \leq \mathbb{E}\left(\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}(g(Z_i)-g(Z_i'))\right) \leq 2\mathbb{E}\left(\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}\eta_i g(Z_i)\right),$$

and, when $a \leq g \leq b$ *and* $2n(b-a)^2 \leq \epsilon^2$,

$$\mathbb{P}[\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}(g(Z_i)-Pg) > \epsilon] \leq 2\mathbb{P}\left(\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}(g(Z_i)-g(Z_i')) > \epsilon/2\right) \leq 4\mathbb{P}\left(\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}\eta_i g(Z_i) > \epsilon/4\right).$$

Exercise 12. *Let* $(Z_1,\ldots,Z_n)$ *be a collection of independent random variables. Let* $(Z_1',\ldots,Z_n')$ *be a random collection with same distribution as* $(Z_1,\ldots,Z_n)$ *and independent from* $(Z_1,\ldots,Z_n)$. *Let* $(\eta_1,\ldots,\eta_n)$ *be a collection of Rademacher variables. Define* $Z_g = \sum_{i=1}^{n}\{g(Z_i)-Eg(Z)\}$ *and* $R_g = \sum_{i=1}^{n}\eta_i\{g(Z_i)-g(Z_i)\}$.

1. *Let* $\Psi : \mathbb{R} \to \mathbb{R}_{\geq 0}$ *increasing and convex. Show that*

$$\mathbb{E}_Z[\Psi(\sup_{g\in\mathcal{G}} Z_g)] \leq \mathbb{E}_Z\mathbb{E}_{Z'}[\Psi(\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}\{g(Z_i)-g(Z_i')\})]$$

2. *Deduce that*

$$\mathbb{E}_Z[\Psi(\sup_{g\in\mathcal{G}} Z_g)] \leq \mathbb{E}_Z\mathbb{E}_\eta[\Psi(2\sup_{g\in\mathcal{G}} R_g)]$$

4 Vapnik-Chervonenkis inequalities

This section focuses on the Vapnik-Chervonenkis (VC) approach for obtaining deviation bounds on suprema of empirical processes. Initially developed in 1971 (see Vapnik and Chervonenkis (2015) for a reprint), this method leverages a combinatorial tool—the VC dimension—to quantify the complexity of a function class and thus control the fluctuation of the associated empirical process.

The VC framework is a powerful illustration of a more general principle: controlling the behavior of empirical processes by measuring the complexity of the underlying function class. In Section 5, we will see that this is part of a broader theory based on metric entropy and covering numbers, with the VC dimension providing one way to bound these quantities.

4.1 A Glivenko-Cantelli theorem

One particularly interesting and relevant estimation problem is the one of the cumulative distribution function. Our goal here is to assess the estimation error (in terms of uniform norm) with the help of a non-asymptotic deviation bound. Let $Z_1, \dots, Z_n$ be an independent collection of random variables with cumulative distribution function F defined on $\mathbb{R}$. Define the estimator

$$\hat{F}(y) = n^{-1} \sum_{i=1}^{n} 1_{Z_i \leq y}.$$

Remark that

$$\sup_{y \in \mathbb{R}} n|\hat{F}(y) - F(y)| = \sup_{g \in \mathcal{G}} |Z_g|$$

with the element of $\mathcal{G}$ being the functions $g_y(z) = 1_{z \leq y}$. By the symmetrization of the probability in Lemma 12, one has

$$\mathbb{P}(\sup_{y \in \mathbb{R}} n|\hat{F}(y) - F(y)| > \epsilon) \leq 4\mathbb{P}\left(\sup_{y \in \mathbb{R}} \left| \sum_{i=1}^{n} \eta_i 1_{Z_i \leq y} \right| > \epsilon/4 \right)$$

The number of vectors in $\{0,1\}^n$ that can be written as $v = (g_y(Z_1), \dots, g_y(Z_1))$ is less than $(n+1)$. Call this set of vector $\mathcal{V}$. We have

$$\mathbb{P}\left(\sup_{y \in \mathbb{R}} \left| \sum_{i=1}^{n} \eta_i 1_{Z_i \leq y} \right| > \epsilon/4 \right) \leq \sum_{(v_1, \dots, v_n) \in \mathcal{V}} \mathbb{P}\left(\left| \sum_{i=1}^{n} \eta_i v_i \right| > \epsilon/4 \right).$$

Since, for any v_i, each $\eta_i v_i$ is sub-Gaussian with factor v_i, we have that $\sum_{i=1}^{n} \eta_i v_i$ is sub-Gaussian with factor $\sum_i v_i$ that is less than n. It follows that

$$\mathbb{P}\left(\sup_{y \in \mathbb{R}} \left| \sum_{i=1}^{n} \eta_i 1_{Z_i \leq y} \right| > \epsilon/2 \right) \leq 2(n+1) \exp(-\epsilon^2/(32n)).$$

We therefore have established the following result, that is often referred to as Glivenko-Cantelli's type theorems.

Theorem 18. *Let $Z_1, \dots, Z_n$ be an independent collection of random variables with cumulative distribution function F defined on $\mathbb{R}$. We have*

$$\mathbb{P}\left(\sup_{y \in \mathbb{R}} |\hat{F}(y) - F(y)| > t \right) \leq 8(n+1) \exp(-t^2 n/32).$$

4.2 Vapnik-Chervonenkis dimension

Let $\mathcal{A}$ be a class of sets in S. Given an arbitrary collection $z = z_1, \ldots, z_n$ of distinct points in S, consider the collection of $\mathbb{R}^n$-points

$$\mathbb{1}_{\mathcal{A}}(z) = \{(\mathbb{1}_A(z_1) \ldots, \mathbb{1}_A(z_n)) : A \in \mathcal{A}\} \subset \{0,1\}^n$$

We have that $|\mathbb{1}_{\mathcal{A}}(z)| \leq 2^n$ and when $|\mathbb{1}_{\mathcal{A}}(z)| = 2^n$ we say that z is shattered by $\mathcal{A}$. An important quantity is therefore

$$\mathbb{S}_n(\mathcal{A}) := \sup_{z \in S^n} |\mathbb{1}_{\mathcal{A}}(z)|$$

which is called the shattering coefficient. It represents loosely speaking the number of possible realization in $\mathbb{1}_{\mathcal{A}}$. The Vapnik dimension is defined as

$$vc(\mathcal{A}) = \max\{n \geq 1 : \exists z \in \mathcal{Z}^n \text{ such that } |\mathbb{1}_{\mathcal{A}}(z)| = 2^n\} = \max\{n \geq 1 : \mathbb{S}_n(\mathcal{A}) = 2^n\}.$$

As a consequence, the fact that all given $z_1, \cdots, z_v$ points cannot be shattered is equivalent to the VC dimension is smaller than v. The reason why the VC dimension is appropriate to control the complexity of classes of sets is perhaps explained by the Sauer's lemma (see Lugosi (2002) for a proof) which states that

$$\mathbb{S}_n(\mathcal{A}) \leq \sum_{i=0}^{vc(\mathcal{A})} \binom{n}{i}$$

Note that when $n \leq vc(\mathcal{A})$ the bound $2^n \leq 2^{vc(\mathcal{A})}$ is better than the above. An interesting consequence is that

$$\mathbb{S}_n(\mathcal{A}) \leq (n+1)^{vc(\mathcal{A})}.$$

Hence the above bound for n large, improves upon the 2^n bound that is given from the definition. Consequently, the number of functions within $\mathbb{1}_{\mathcal{A}}(z)$ is of order $n^{vc(\mathcal{A})}$ and, loosely speaking, suprema mith be replaced by maximum. Two examples are now given to illustrate the previous. The class of cells $(-\infty, t]$ can be used to shatter any set with 1 point but fails to shatter any set with 2 points. Its Vapnik dimension is therefore 1. The class of cells $(s, t]$ can be used to shatter any set with 2 distinct points but fails to shatter any set with 3 distinct points. Its Vapnik dimension is therefore 2. An elementary preservation property is now given.

Lemma 19. *We have that $vc(\mathcal{A}) = vc(\mathcal{A}^c)$ and whenever $\mathcal{A} \subset \mathcal{B}$, $vc(\mathcal{A}) \leq vc(\mathcal{B})$.*

4.3 Main inequalities

We now give the first Vapnik's inequality, tailored to functions valued in $\{0,1\}$ with finite Vapnik dimension.

Theorem 20 (First Vapnik inequality)**.** *Under (Ziid), for any Borelian class $\mathcal{A} \subset \mathcal{S}$, $\delta > 0$ and $n \geq 1$, it holds with probability at least $1 - \delta$:*

$$\left| \sum_{i=1}^{n} (\mathbb{1}_B(Z_i) - P(B)) \right| \leq \sqrt{8n \log(4\mathbb{S}_{\mathcal{A}}(2n)/\delta)}, \qquad \forall B \in \mathcal{A}.$$

Proof. Let $Z_B = |\sum_{i=1}^n (\mathbb{1}_B(Z_i) - P(B))|$ and $R_B = |\sum_{i=1}^n \eta(\mathbb{1}_B(Z_i) - \mathbb{1}_B(Z_i'))|$. We have, whenever $2n \leq t^2$,

$$\mathbb{P}(Z_{\mathcal{A}} > t) \leq 2\mathbb{P}(R_{\mathcal{A}} > t/2)$$

But each R_B depends on $(1_B(Z_1), \ldots, 1_B(Z_n))$ and $(1_B(Z_1'), \ldots, 1_B(Z_n'))$. The number of elements that are described when changing B is at most $\mathbb{S}_{\mathcal{A}}(2n)$. As a consequence,

$$\mathbb{P}_\eta(R_{\mathcal{A}} > t/2) \leq \mathbb{S}_{\mathcal{A}}(2n) \max_{B \in \mathcal{A}} \mathbb{P}_\eta(R_B > t/2).$$

But, since each R_B is sub-Gaussian with factor n, we obtain

$$\mathbb{P}(Z_{\mathcal{A}} > t) \leq 4\mathbb{S}_{\mathcal{A}}(2n) \exp(-(t/2)^2/2n).$$

To conclude, note that when $t^2 < 2n$, then $4\exp(-t^2/(8n)) > 4\exp(-1/4) \geq 1$ so the above bound is valid for all $t > 0$ and $n \geq 1$. Taking t to achieve the right probability level leads to the stated result. □

The above proof of Vapnik's inequality is based on a symmetrization of probability, stated in Lemma 12. Another proof is investigated in Exercise 14. The obtained result turns out to be better, as we get, with probability at least $1 - \delta$:

$$\left| \sum_{i=1}^n (\mathbb{1}_B(Z_i) - P(B)) \right| \leq \sqrt{8n \log(2\mathbb{S}_{\mathcal{A}}(n)/\delta)}, \qquad \forall B \in \mathcal{A}.$$

The previous inequality fails to be tight when a class of function has small variance. The next inequality allows to include a variance term in the bound. This result next result is established in (Anthony and Shawe-Taylor, 1993, Theorem 2.1) (see also Theorem 1.11 in Lugosi (2002)). Other related results include Theorem 15 in Chaudhuri and Dasgupta (2010); Theorem 1 in Goix et al. (2015) and more recently Corollary 4.4 in Lhaut et al. (2022).

Theorem 21 (Second Vapnik inequality)**.** *Under (Ziid), for any Borelian class $\mathcal{A} \subset \mathcal{S}$, $\delta > 0$ and $n \geq 1$, it holds with probability at least $1 - \delta$:*

$$\sum_{i=1}^n (\mathbb{1}_B(X_i) - P(B)) \geq -\sqrt{4nP(B) \log(4\mathbb{S}_{\mathcal{A}}(2n)/\delta)}, \qquad \forall B \in \mathcal{A},$$

Proof. The proof relies on the second symmetrization of probability as exposed in Lemma 14. Then one can introduce Rademacher variables using that the collection (g_i, g_i') has same distribution as the collection $1_{\{\eta_i=1\}}(g_i, g_i') + 1_{\{\eta_i=-1\}}(g_i', g_i)$ and conclude applying the sub-Gaussian bound. □

The previous results are interesting when the Vapnik dimension $vc(\mathcal{A})$ (introduced in Section 6.3.3) is finite so that $\mathbb{S}_{\mathcal{A}}(n) \leq (n+1)^{vc(\mathcal{A})}$. In Devroye et al. (2013), Lemma 29.1, it shown that the previous concepts can be extended to more general function classes using the class generated by sub-graphs, a notion that will be useful in dealing with the kernel smoothing estimator of the density.

4.4 Examples

Many popular classes of sets have a known Vapnik dimension. Some relevant results might be found in Wenocur and Dudley (1981). For instance, it is shown that the class of cells made of elements as $(-\infty, b] = \prod_{k=1}^{d}(-\infty, b_k]$, where $b \in \mathbb{R}^d$ has a Vapnik dimension equal to d. This is specified in the next lemma.

Lemma 22. *We have* $vc(\{(-\infty, b] : b \in \mathbb{R}^d\}) = d$ *and* $vc(\{(-a, b] : b \in \mathbb{R}^d, a \in \mathbb{R}^d\}) = 2d$.

Similarly, the class of balls has Vapnik dimension equal to $d + 1$ as stated below.

Lemma 23. *We have* $vc(\{B(x, r) : x \in \mathbb{R}^d, r > 0\}) = d + 1$.

We also have the following lemma about hyperplanes.

Lemma 24. *We have* $vc(\{\langle w, x\rangle > 0 : w \in \mathbb{R}^d\}) = d$.

A proof of a slightly more general result is given in (Giné and Nickl, 2021), Proposition 3.6.6. An important consequence is for *support vector machine* classification because the previous result implies that the class of all affine hyperplanes in $\mathbb{R}^d$ is of dimension $d + 1$.

4.5 An estimate on the supremum of the nearest neighbors radius

We use the notation from Section 2.3. Aim is to obtain a bound on the k-NN radius $\hat{\tau}_{n,k,x}$ that holds uniformly in x on some set. This is a stronger result than Lemma 8 which is concerned with a particular value of x. Similar assumptions are needed except that k needs to be slightly larger.

(X) The distribution of X admits a density f on $\mathbb{R}^d$. Let $K \subset \mathbb{R}^d$. There is $\tau_0 > 0$ such that $b := \inf_{y \in K} f(y) > 0$ and for all $\tau \leq \tau_0$, $\int_{K \cap B(x,\tau_0)} \geq c \int_{B(x,\tau_0)}$.

Lemma 25. *Suppose (H) and (X) are fulfilled. Let* $n \geq 1$, $1 \leq k \leq n$ *and* $\delta \in (0, 1)$, *be such that* $16d \log(12n/\delta) \leq k \leq \tau_0^d nbcV_d/2$. *We have with probability* $1 - \delta$,

$$\sup_{x \in K} \hat{\tau}_{n,k,x} \leq \left(\frac{2k}{nbcV_d}\right)^{1/d}.$$

Proof. The proof is almost the same as that of Lemma 8. The second Vapnik inequality will be used instead of the Chernoff multiplicative bound. Let

$$\overline{\tau}_{n,k} = \left(\frac{2k}{nbcV_d}\right)^{1/d}$$

It holds for all $x \in K$,

$$P(B(x, \overline{\tau}_{n,k}) \geq 2k/n.$$

But using Theorem 21, with probability $1 - \delta$, for all x,

$$nP_nB(x, \overline{\tau}_{n,k} \geq nP(B(x, \overline{\tau}_{n,k})) - \sqrt{4nP(B(x, \overline{\tau}_{n,k})) \log(4\mathbb{S}_{\mathcal{B}}(2n)/\delta)},$$

where $\mathcal{B} = \{B(x, r) : x \in \mathbb{R}^d, r > 0\})$. Since $S_{\mathcal{B}}(n) \leq (n + 1)^V$ where V is the Vapnik dimension of $\mathcal{B}$, and since the set of closed balls has a Vapnik dimension equal to $d + 1$ as claimed

before, we find that $S_{\mathcal{B}}(2n) \leq (2n+1)^{d+1}$. But $4\log(4\mathbb{S}_{\mathcal{A}}(2n)/\delta) \leq 4\log(4(2n+1)^{d+1}/\delta) \leq 4\log(4(3n)^{d+1}/\delta) \leq 4(d+1)\log(12n/\delta) \leq 8d\log(12n/\delta)$ and we find

$$nP_nB(x,\overline{\tau}_{n,k} \geq nP(B(x,\overline{\tau}_{n,k})) - \sqrt{8dnP(B(x,\overline{\tau}_{n,k}))\log(12n/\delta)},$$

The function $x \mapsto x - \sqrt{4xl}$ being increasing when $x \geq l$ we get whenever $nP(B(x,\overline{\tau}_{n,k})) \geq 2\log(12n/\delta)$, and, sufficiently, whenever $2k \geq 2\log(12n/\delta)$, that for all x,

$$nP_nB(x,\overline{\tau}_{n,k}) \geq 2k - \sqrt{16k\log(12n/\delta)}$$

and if $k \geq 16\log(12n/\delta)$ we obtain, with probability at least $1-\delta$, that, for all $x \in K$, $nP_nB(x,\overline{\tau}_{n,k}) \geq k$ which allows to conclude.

□

4.6 Exercises

Exercise 13. *Suppose that all elements $g \in \mathcal{G}$ are such that $0 \leq g(x) \leq U$. Show that*

$$\mathbb{E}[\sup_{g\in\mathcal{G}} n^{-1}\sum_{i=1}^{n}\{g(Z_i) - P(g(Z))\}] \leq 2U\mathbb{E}[\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}\eta_i \mathbb{1}_{\{U_i<g(Z_i)\}}]$$

for some collection $(U_i)_{1\leq i\leq n}$ of independent and identically distributed random variables. Give a Vapnik inequality (of the first type) for such classes.

Exercise 14 (improved Vapnik's inequality)**.** *Let $(Z_1,\ldots,Z_n)$ be a collection of independent random variables. Define $Z_g = \sum_{i=1}^{n}\{g(Z_i) - Eg(Z)\}$.*

1. *Let $(Z_1',\ldots,Z_n')$ be a random collection with same distribution as $(Z_1,\ldots,Z_n)$ and independent from $(Z_1,\ldots,Z_n)$ and $(\eta_1,\ldots,\eta_n)$. Let $\Psi : \mathbb{R} \to \mathbb{R}_{\geq 0}$ non-decreasing and convex. Show that*

$$\mathbb{E}_Z[\Psi(\sup_{g\in\mathcal{G}} Z_g)] \leq \mathbb{E}_Z\mathbb{E}_{Z'}[\Psi(\sup_{g\in\mathcal{G}}\sum_{i=1}^{n}\{g(Z_i) - g(Z_i')\})].$$

2. *Deduce that*

$$\mathbb{E}_Z[\Psi(\sup_{g\in\mathcal{G}} Z_g)] \leq \mathbb{E}_Z\mathbb{E}_\eta[\Psi(2\sup_{g\in\mathcal{G}} R_g)].$$

3. *Suppose that $g = \mathbb{1}_A$ where $A \in \mathcal{A}$, is a collection of measurable sets in S. Let $z = (z_1,\ldots,z_n) \subset S$, $\mathcal{W}(z) = \{(\mathbb{1}_A(z_1),\ldots,\mathbb{1}_A(z_n)) : A \in \mathcal{A}\}$. Show that*

$$\mathbb{E}_\eta[\Psi(2\sup_{g\in\mathcal{G}} R_g)] \leq \sum_{(w_1,\ldots,w_n)\in\mathcal{W}(z)} \mathbb{E}_\eta[\Psi(2\sum_{i=1}^{n}\eta_i w_i)].$$

4. *Let $\mathbb{S}_n(\mathcal{A}) = \max_{z_1,\ldots,z_n} |\mathcal{W}(z)|$ where for any collection of sets A, $|A|$ denotes the number of elements in A. Show that*

$$\mathbb{P}(\sup_{g\in\mathcal{G}} Z_g > t) \leq \mathbb{S}_n(\mathcal{A})\exp(-t^2/(8s^2))$$

   ($\Psi := \Psi_{\lambda,t}$ shall be well chosen)

5. *Show that*

$$\mathbb{P}(\sup_{g\in\mathcal{G}} |Z_g| > t) \leq 2\mathbb{S}_n(\mathcal{A})\exp(-t^2/(8s^2))$$

5 Covering numbers and Dudley's entropy integral

The objective of the section is to present Dudley's entropy integral bound on the expectation of the rademacher sums:

$$\mathbb{E}_\eta[\sup_{g\in\mathcal{G}} \sum_{i=1}^n \eta_i g(Z_i)].$$

We start considering the case where $\mathcal{G}$ is finite. Then, after introduction covering numbers, we derive Dudley's result using a chaining argument.

5.1 Finite classes

We have the following lemma which will be crucial to obtain an upper bound on the Rademacher complexity.

Lemma 26. *For each $g \in \mathcal{G}$ and any collection $(z_1, \ldots, z_n) \subset S$, the random variable $\sum_{i=1}^n \eta_i g(z_i)$ is sub-Gaussian with factor $\sum_{i=1}^n g(z_i)^2$.*

Proof. By Hoeffding Lemma (see Exercise 3), it holds that $[a, b]$-valued random variables are subGaussian with factor $(b-a)^2/4$. It follows that $\eta_i g(z_i)$ is sub-Gaussian with factor $g(z_i)^2$. The result follows using that sum of sub-Gaussian is still sub-Gaussian. □

Before introducing Vapnik-Chervonenkis dimension, let us give a basic upper-bound on the Rademacher complexity that is valid when the class is made of a finite number of functions. Let us introduce

$$\|g\|^2_{L_2(P_n)} = n^{-1} \sum_{i=1}^n g(Z_i)^2,$$

as well as

$$\hat{\sigma}_n^2(\mathcal{G}) = \sup_{g\in\mathcal{G}} \|g\|^2_{L_2(P_n)}.$$

Lemma 27. *For any collection $(z_1, \ldots, z_n) \subset S$, and a class $\mathcal{G}$ having cardinality $M \geq 1$. It holds with probability 1,*

$$\mathbb{E}_\eta[\sup_{g\in\mathcal{G}} \sum_{i=1}^n \eta_i g(Z_i)] \leq \sqrt{2n\hat{\sigma}_n^2(\mathcal{G}) \log(M)}$$
$$\mathbb{E}_\eta[\sup_{g\in\mathcal{G}} |\sum_{i=1}^n \eta_i g(Z_i)|] \leq \sqrt{2n\hat{\sigma}_n^2(\mathcal{G}) \log(2M)}.$$

Proof. The first inequality follows from Lemma 7 and Lemma 26. The second bound can be obtained using either that $\exp(|x|) \leq \exp(x) + \exp(-x)$ and adapting Lemma 7 or simply remarking that $|x| = \max(x, -x)$ so that

$$|R_g| = \max(R_g, -R_g) \leq \sup_{g\in\{\mathcal{G}\cup-\mathcal{G}\}} R_g$$

where $R_g = \sum_{i=1}^n \eta_i g(z_i)$. Then we conclude applying the first statement.

□

5.2 Covering numbers

A standard way to control the fluctuation of the Rademacher complexity relies on the concept of covering numbers. Those numbers are useful to measure the size or the complexity of a function class. Given $\epsilon > 0$, the ϵ-covering number is the smallest number of closed balls of radius $\epsilon > 0$ required to cover $\mathcal{G}$. The smaller the covering numbers the more simple the function class and vice-versa. The next definition is formal and relies on ϵ-net.

Definition 28 (ϵ-net and covering numbers)**.** *Given a metric space $(\mathcal{G}, d)$, a set $\mathcal{G}_\epsilon$ is called ϵ-net if for all $g \in \mathcal{G}$ there is $\pi(g) \in \mathcal{G}_\epsilon$ such that $d(g, \pi(g)) \leq \epsilon$. The ϵ-covering number, denoted $\mathcal{N}(\mathcal{G}, d, \epsilon)$, is defined as the smallest cardinality of an ϵ-net.*

We therefore focus on classes $(\mathcal{G}, d)$ that are called totally bounded, i.e., for any $\epsilon > 0$, there exist finitely many balls of radius ϵ that can be used to cover $\mathcal{G}$. In the application we have in mind the space $\mathcal{G}$ is a subsapce of $L_2(Q)$ and an ϵ-net don't have to be included in $\mathcal{G}$ but just to be elements in $L_2(Q)$. Here are some preservation properties related to covering numbers.

Lemma 29. *Suppose that $\mathcal{F}$ and $\mathcal{G}$ have envelop $U_\mathcal{F}$ and $U_\mathcal{G}$. Then*

$$\begin{aligned}
&\mathcal{N}(-\mathcal{F}, L_2(Q), \epsilon U_\mathcal{F}) \leq \mathcal{N}(\mathcal{F}, L_2(Q), \epsilon U_\mathcal{F}) \\
&\mathcal{N}(\mathcal{F} \cup \mathcal{G}, L_2(Q), \epsilon U_\mathcal{F} \vee U_\mathcal{G}) \leq \mathcal{N}(\mathcal{F}, L_2(Q), \epsilon U_\mathcal{F}) + \mathcal{N}(\mathcal{G}, L_2(Q), \epsilon U_\mathcal{G}) \\
&\mathcal{N}(\mathcal{F} + \mathcal{G}, L_2(Q), \epsilon(U_\mathcal{F} + U_\mathcal{G})) \leq \mathcal{N}(\mathcal{F}, L_2(Q), \epsilon U_\mathcal{F}) \mathcal{N}(\mathcal{G}, L_2(Q), \epsilon U_\mathcal{G}) \\
&\mathcal{N}(\mathcal{F}\mathcal{G}, L_2(Q), \epsilon U_\mathcal{F} U_\mathcal{G}) \leq \mathcal{N}(\mathcal{F}, L_2(Q), \epsilon U_\mathcal{F}/2) \mathcal{N}(\mathcal{G}, L_2(Q), \epsilon U_\mathcal{G}/2)
\end{aligned}$$

5.3 Main results

The following result, due to Dudley (see Van Handel (2014), Corollary 5.25 or Boucheron et al. (2013), Corollary 13.2, or (Zhang, 2023)) is crucial in the empirical process theory. It consists in the following upper-bound on the Rademacher complexity using the entropy-integral. The proof relies on the previous bound (for finite class) and the so-called chaining method.

Theorem 30 (Dudley)**.** *For any collection $(Z_1, \ldots, Z_n) \subset S$, it holds with probability 1,*

$$\mathbb{E}_\eta[\sup_{g \in \mathcal{G}} \sum_{i=1}^n \eta_i g(Z_i)] \leq 12\sqrt{n} \int_0^{\hat{\sigma}_n(\mathcal{G})/2} \sqrt{\log(\mathcal{N}(\mathcal{G}, L_2(P_n), \epsilon))} \mathrm{d}\epsilon.$$

Moreover, we have, with probability 1,

$$\mathbb{E}_\eta[\sup_{g \in \mathcal{G}} \left| \sum_{i=1}^n \eta_i g(Z_i) \right|] \leq 12\sqrt{n} \int_0^{\hat{\sigma}_n(\mathcal{G})/2} \sqrt{\log(2\mathcal{N}(\mathcal{G}, L_2(P_n), \epsilon))} \mathrm{d}\epsilon.$$

Proof. The proof given here is adapted from (Zhang, 2023) and from S. Kakade's lecture notes (spring 2008). Set $\mathcal{G}_0 = \{0\}$ and $\epsilon_0 = \hat{\sigma}_n(\mathcal{G})$. Let $\epsilon_k = 2^{-k}\epsilon_0$, $k = 1, 2, \ldots$ and let $\mathcal{G}_k$ be a ϵ_k-net. Write

$$g = g - \pi_L(g) + \sum_{k=1}^L (\pi_k(g) - \pi_{k-1}(g))$$

Triangle inequality gives $\|\pi_k(g) - \pi_{k-1}(g)\|_{L_2(P_n)} \leq 3\epsilon_k$. As a consequence of Lemma 27, we have

$$\begin{aligned}
\mathbb{E} \sup_{g \in \mathcal{G}} \sum_{i=1}^n \eta_i (\pi_k(g) - \pi_{k-1}(g))_i &\leq 3\epsilon_k \sqrt{2n \log(N_k N_{k-1})} \\
&\leq 6(\epsilon_k - \epsilon_{k+1}) \sqrt{2n \log(N_k N_{k-1})} \\
&\leq 12(\epsilon_k - \epsilon_{k+1}) \sqrt{n \log(N_k)} \\
&= 12\sqrt{n \log(N_k)} \int_{\epsilon_{k+1}}^{\epsilon_k} du \\
&\leq 12\sqrt{n} \int_{\epsilon_{k+1}}^{\epsilon_k} \sqrt{\log(N(\mathcal{G}, L_2(P_n), u))} du
\end{aligned}$$

Taking the sum over k we get that

$$\mathbb{E} \sum_{k=1}^L \sup_{g \in \mathcal{G}} \sum_{i=1}^n \eta_i (\pi_k(g) - \pi_{k-1}(g))_i \leq 12\sqrt{n} \int_{\epsilon_L/2}^{\epsilon_1} \sqrt{\log(N(\mathcal{G}, L_2(P_n), u))} du.$$

Considering the first term, one has that (from Cauchy-Schwarz inequality)

$$|\sum_{i=1}^n \eta_i (g - \pi_L(g))_i| \leq n \|g - \pi_L(g)\|_{L_2(P_n)} \leq n\epsilon_L.$$

Consequently,

$$\mathbb{E}_\eta [\sup_{g \in \mathcal{G}} \sum_{i=1}^n \eta_i g(Z_i)] \leq n\epsilon_L + 12\sqrt{n} \int_0^{\epsilon_0/2} \sqrt{\log(N(\mathcal{G}, L_2(P_n), u))} du$$

Taking the limit with respect to $L \to \infty$ leads to the result.

Now we consider the second statement. Set $R_g = \sum_{i=1}^n \eta_i g(Z_i)$. We have

$$|R_g| \leq R_g \vee -R_g \leq \sup_g R_g \vee \sup_g -R_g.$$

Noting that $\sup_g R_g$ and $\sup_g -R_g$ are both smaller than $\sup_{g \in \mathcal{G} \cup -\mathcal{G}} R_g$. We obtain that

$$\sup_g |R_g| \leq \sup_{g \in \mathcal{G} \cup -\mathcal{G}} R_g.$$

(one can also show the reverse inequality). Now remark that covering numbers of $\mathcal{G} \cup -\mathcal{G}$ are smaller than twice the covering numbers of $\mathcal{G}$. We conclude by applying Theorem 30 with the class $\mathcal{G} \cup -\mathcal{G}$. □

The distance in $L_2(P_n)$ that appears in the previous statement is therefore important and we shall need at some point to have an upper bound on $\hat{\sigma}_n^2(\mathcal{G})$. The following is an easy consequence of the contraction principle given in the last statement of Lemma 15.

Lemma 31. *Under (Ziid), for any class $\mathcal{G}$ with envelop U, we have*

$$n\mathbb{E}\hat{\sigma}_n^2(\mathcal{G}) \leq n \sup_{g \in \mathcal{G}} P(g^2) + 4U\mathbb{E}[\sup_{g \in \mathcal{G}} \sum_{i=1}^n \eta_i g(Z_i)]$$

The previous result can be found in Corollary 3.4 in Talagrand (1994) (with a less accurate constant 8 rather than the above 4) or in Van Handel (2014). The proof is the subject of an exercise in next section.

5.4 Exercises

Exercise 15. *Define* $\tilde{\sigma}_n^2(\mathcal{G}) = n^{-1}\sup_{g\in\mathcal{G}}\sum_{i=1}^n(g(Z_i)^2 - \mathbb{P}g^2(Z))$. *Under the assumption of Lemma 31 show that*

$$n\mathbb{E}\tilde{\sigma}_n^2(\mathcal{G}) \leq 4U\mathbb{E}[\sup_{g\in\mathcal{G}}\sum_{i=1}^n \eta_i g(Z_i)].$$

Conclude by showing the statement of Lemma 31.

Exercise 16. *Suppose that* $\mathcal{G}$ *has envelop function* G. *Show that for any probability measure* Q,

$$\mathcal{N}(\mathcal{G}^2, L_2(Q), \epsilon\|2G^2\|_{L_2(Q)}) \leq \mathcal{N}(\mathcal{G}, L_2(\tilde{Q}), \epsilon\|G\|_{L_2(\tilde{Q})})$$

where $d\tilde{Q} \propto (2G)^2 dQ$.

Exercise 17. *Suppose that, for all* $\epsilon \in (0,1)$,

$$\mathcal{N}(\mathcal{G}, L_2(Q), \epsilon U) \leq (A/\epsilon)^v$$

where (A, v) *are two constants larger than* 1. *Define* $\tilde{\sigma}_n^2(\mathcal{G}) = n^{-1}\sup_{g\in\mathcal{G}}\sum_{i=1}^n(g(Z_i)^2 - \mathbb{P}g^2(Z))$. *Show that*

$$n\mathbb{E}\tilde{\sigma}_n^2(\mathcal{G}) \leq 24\sqrt{n}\mathbb{E}\int_0^{\hat{\sigma}_n(\mathcal{G}^2)/2}\sqrt{\log(\mathcal{N}(\mathcal{G}^2, L_2(P_n), \epsilon))}\mathrm{d}\epsilon$$

Conclude that

$$n\mathbb{E}\tilde{\sigma}_n^2(\mathcal{G}) \leq 12U^2\sqrt{2nv\log(4eA)}$$

Exercise 18. *Show that*

$$\begin{aligned}&\mathbb{E}_\eta[\sup_{g\in\mathcal{G}}\sum_{i=1}^n \eta_i(g(Z_i) - g(Z_i'))]\\ &\leq 12\sqrt{n}\int_0^{(s_n(\mathcal{G})+s_n'(\mathcal{G}))/2}\sqrt{\log(\mathcal{N}(\mathcal{G}, L_2(P_n), \epsilon/2)\mathcal{N}(\mathcal{G}, L_2(P_n'), \epsilon/2))}\mathrm{d}\epsilon\end{aligned}$$

where $s_n^2(\mathcal{G}) = n^{-1}\sup_{g\in\mathcal{G}}\sum_{i=1}^n(g(Z_i) - \mathbb{E}g(Z))^2$ *and* P_n *is empirical measure on* $(Z_i)_{1\leq i\leq n}$ *(*s_n' *and* P_n' *are defined accordingly with respect to* Z_i'*).*

Exercise 19. *Let* $Lip = \{f : [0,1] \to [0,1] : |f(x) - f(y)| \leq |x - y|\}$. *Recall that* $W_1(P_n, P) = \sup_{f\in Lip} P_n(f) - P(f)$ *and that* $\mathcal{N}(Lip, \|\cdot\|_\infty, \epsilon) \leq \exp(c/\epsilon)$. *Show that* $\mathbb{E}[W_1(P_n, P)] \leq 24\sqrt{2nc}$ *and conclude that* $W_1(P_n, P) \to 0$ *almost surely.*

6 VC classes

6.1 Definition

The concept of VC class is standard to obtain uniform bounds on the empirical process. While different definitions exist we here follow the definition given in Nolan and Pollard (1987); Giné and Guillou (2002) which is based on uniform entropy numbers. This notion is more general than the one initially introduced in Vapnik and Chervonenkis (2015) as detailed for instance in (Van Der Vaart and Wellner, 1996, Theorem 2.6.4) and as shown next.

Given a probability measure Q on $(S, \mathcal{S})$, the metric space of squared-integrable functions with respect to Q is defined as

$$L_2(Q) = \left\{ g : S \mapsto \mathbb{R} \text{ such that } \|g\|_{L_2(Q)}^2 := Q(g^2) < \infty \right\}.$$

Definition 32. *A class $\mathcal{G}$ of pointwise measurable (Van Der Vaart and Wellner, 1996, Example 2.3.4) real-valued functions on a measurable space $(S, \mathcal{S})$ is said to be VC with parameters $(v, A) \in (1, \infty) \times [e, \infty)$ and envelope G if for any $0 < \epsilon < 1$ and any probability measure Q on $(S, \mathcal{S})$, we have*

$$\mathcal{N}\left(\mathcal{G}, L_2(Q), \epsilon \|G\|_{L_2(Q)}\right) \leq (A/\epsilon)^v.$$

The condition on the parameter (v, A) might be alleviated at the price of more technicalities in the proof of the main results. Our choice $A \geq e$ and $v \geq 1$ allows to keep the constant reasonable while maintaining clarity in the proof.

6.2 Main results

The previous inequality can be applied for bounding the Rademacher complexity associated with function classes that are localized in terms of variance. That is, we define

$$\mathcal{G}_r = \{g \in \mathcal{G} \,:\, P_n(g^2) \leq r\},$$

and introduce the notation

$$\begin{aligned}
\mathrm{Rad}_{n,s}(\mathcal{F}) &= \mathbb{E}_\eta[\sup_{g \in \mathcal{F}} \sum_{i=1}^n \eta_i g(Z_i)], \\
\mathrm{Rad}_n(\mathcal{F}) &= \mathbb{E}_\eta[\sup_{g \in \mathcal{F}} \left| \sum_{i=1}^n \eta_i g(Z_i) \right|].
\end{aligned}$$

When $r > U^2$, since $P_n(g^2) \leq U^2$, we have that $\mathcal{G}_r = \mathcal{G}_{U^2}$. Therefore we assume subsequently that $r \leq U^2$.

Theorem 33. *Suppose that $\mathcal{G}$ is of VC type with envelop U and parameter (v, A). For any collection $(Z_1, \ldots, Z_n) \subset S$ and any $r \leq U^2$, it holds that, with probability 1,*

$$\mathrm{Rad}_{n,s}(\mathcal{G}_r) \leq 8\sqrt{nrv \log(2AU/\sqrt{r})}.$$

In particular, using $r = \sup_{g \in \mathcal{G}} P_n(g^2) =: \hat{\sigma}_n^2(\mathcal{G})$ and $r = U^2$, we find, respectively,

$$\begin{aligned}
\mathrm{Rad}_{n,s}(\mathcal{G}) &\leq 8\sqrt{\hat{\sigma}_n^2(\mathcal{G}) nv \log(2AU/\hat{\sigma}_n(\mathcal{G}))} \\
\mathrm{Rad}_{n,s}(\mathcal{G}) &\leq 8\sqrt{U^2 nv \log(2A)}.
\end{aligned}$$

Proof. Applying Theorem 30, we find

$$\mathrm{Rad}_{n,s}(\mathcal{G}_r) \leq 12\sqrt{n}\int_0^{\sqrt{r}/2}\sqrt{\log(\mathcal{N}(\mathcal{G}, L_2(P_n), \epsilon))}\mathrm{d}\epsilon.$$

An envelop for the class is U hence

$$\mathrm{Rad}_{n,s}(\mathcal{G}_r) \leq 12U\sqrt{n}\int_0^{\sqrt{r}/(2U)}\sqrt{\log(\mathcal{N}(\mathcal{G}, L_2(P_n), U\epsilon))}\mathrm{d}\epsilon$$

Using Definition 32, we obtain

$$\mathrm{Rad}_{n,s}(\mathcal{G}_r) \leq 12U\sqrt{vn}\int_0^{\sqrt{r}/(2U)}\sqrt{\log(A/\epsilon)}\mathrm{d}\epsilon$$

By change of variable $A/\epsilon = u$, it follows that

$$\mathrm{Rad}_{n,s}(\mathcal{G}_r) \leq 12AU\sqrt{vn}\int_{2AU/\sqrt{r}}^{+\infty} u^{-2}\sqrt{\log(u)}\mathrm{d}u.$$

For all $t \geq 1$, integration by parts gives

$$\begin{aligned}
\int_t^{+\infty} u^{-2}\sqrt{\log(u)}\mathrm{d}u &= [-u^{-1}\sqrt{\log(u)}]_t^{+\infty} + (1/2)\int_t^{+\infty} u^{-2}\log(u)^{-1/2}\mathrm{d}u \\
&\leq t^{-1}\sqrt{\log(t)} + (1/2)\log(t)^{-1/2}\int_t^{+\infty} u^{-2}\mathrm{d}u \\
&= t^{-1}\sqrt{\log(t)} + (1/2)\log(t)^{-1/2}[-u^{-1}]_t^{+\infty} \\
&\leq t^{-1}\sqrt{\log(t)}(1 + (2\log(t))^{-1}).
\end{aligned}$$

Hence, since $2AU/\sqrt{r} \geq 1$,

$$\int_{2AU/\sqrt{r}}^{+\infty} u^{-2}\sqrt{\log(u)}\mathrm{d}u \leq (2AU/\sqrt{r})^{-1}\sqrt{\log(2AU/\sqrt{r})}(1 + (2\log(2AU/\sqrt{r}))^{-1})$$

leading to

$$\mathrm{Rad}_{n,s}(\mathcal{G}_r) \leq 6\sqrt{nrv\log(2AU/\sqrt{r})}(1 + (2\log(2AU/\sqrt{r}))^{-1})$$

Since $A \geq e$ and $r \leq U^2$, $2AU/\sqrt{r} \geq 2A \geq 2e$, so

$$\mathrm{Rad}_{n,s}(\mathcal{G}_r) \leq 6\sqrt{nrv\log(2AU/\sqrt{r})}(1 + (2\log(2e))^{-1}) \leq 8\sqrt{nrv\log(2AU/\sqrt{r})}.$$

□

A similar assertion holds for Rad_n in place of $\mathrm{Rad}_{n,s}$. This leads to a slight increase of A which becomes $2A$.

Theorem 34. *Suppose that $\mathcal{G}$ is of VC type with envelop U and parameter (v, A). For any collection $(z_1, \ldots, z_n) \subset S$ and any $r \leq U^2$, it holds that, with probability 1,*

$$\mathrm{Rad}_n(\mathcal{G}_r) \leq 8\sqrt{rnv\log(4AU/\sqrt{r})}.$$

In particular, using $r = \sup_{g\in\mathcal{G}} P_n(g^2) =: \hat{\sigma}_n^2(\mathcal{G})$ and $r = U^2$, we find

$$\begin{aligned}
\mathrm{Rad}_n(\mathcal{G}) &\leq 8\sqrt{\hat{\sigma}_n^2(\mathcal{G})nv\log(4AU/\hat{\sigma}_n(\mathcal{G}))} \\
\mathrm{Rad}_n(\mathcal{G}) &\leq 8\sqrt{U^2nv\log(4A)}.
\end{aligned}$$

Proof. For any class $\mathcal{F}$, since $|\sum_i \eta_i g(Z_i)| = \max\big(\sum_i \eta_i g(Z_i), \sum_i \eta_i(-g)(Z_i)\big)$,

$$\mathrm{Rad}_n(\mathcal{F}) = \mathbb{E}_\eta\Big[\sup_{g\in\mathcal{F}}\Big|\sum_i \eta_i g(Z_i)\Big|\Big] = \mathbb{E}_\eta\Big[\sup_{h\in\mathcal{F}\cup-\mathcal{F}}\sum_i \eta_i h(Z_i)\Big] = \mathrm{Rad}_{n,s}(\mathcal{F}\cup-\mathcal{F}).$$

Moreover, since $P_n(g^2) = P_n((-g)^2)$, $(\mathcal{G}\cup-\mathcal{G})_r = \mathcal{G}_r \cup -(\mathcal{G}_r)$, so

$$\mathrm{Rad}_n(\mathcal{G}_r) = \mathrm{Rad}_{n,s}\big((\mathcal{G}\cup-\mathcal{G})_r\big).$$

The class $\mathcal{G}\cup-\mathcal{G}$ has, for any Q and ϵ, covering number

$$\mathcal{N}(\mathcal{G}\cup-\mathcal{G}, L_2(Q), \epsilon\|U\|_{L_2(Q)}) \le 2\,\mathcal{N}(\mathcal{G}, L_2(Q), \epsilon\|U\|_{L_2(Q)}) \le 2(A/\epsilon)^v = (2^{1/v}A/\epsilon)^v \le (2A/\epsilon)^v$$

(using $v \ge 1$), so it is VC type with envelope U and parameter $(v, 2A)$. Applying Theorem 33 to $\mathcal{G}\cup-\mathcal{G}$ (with parameter $2A$ in place of A) gives

$$\mathrm{Rad}_n(\mathcal{G}_r) = \mathrm{Rad}_{n,s}\big((\mathcal{G}\cup-\mathcal{G})_r\big) \;\le\; 8\sqrt{nrv\log\big(2(2A)U/\sqrt{r}\big)} \;=\; 8\sqrt{nrv\log(4AU/\sqrt{r})}. \quad \square$$

Note that the chaining approach allows to reduce the bound that is obtained in the first Vapnik inequality. Indeed, consider the class $\mathcal{G}$ made with indicators of rectangles (shown to be of VC type next). The Vapnik's first inequality bound scales as $\sqrt{nd\log(n)}$ while, following the last inequality from previous theorem, we obtain an upper bound scaling as $\sqrt{nd}$.

One further result, which can be found in Giné and Guillou (2002), with different constants, provides a bound on the expected Rademacher complexity. Note that the next upper bound is similar to the classic Bernstein inequality as we have a variance term and another smaller term that depends on U, the uniform bound on the random variables of interest.

Theorem 35. *Under (Ziid), suppose that $\mathcal{G}$ is of VC type with envelop U and parameter (v, A). Suppose that $\sup_{g\in\mathcal{G}} E(g^2) \le \sigma^2$. It holds that*

$$\mathbb{E}\mathrm{Rad}_{n,s}(\mathcal{G}) \le C\sqrt{vn\sigma^2\log(2AU/\sigma)} + 4C^2Uv\log(2AU/\sigma).$$

with $C = 8$. The same bound is valid for $\mathbb{E}\mathrm{Rad}_n(\mathcal{G})$ but with $4AU$ in place of $2AU$.

Proof. Apply Theorem 33. We get using (twice) Jensen inequality (functions $\sqrt{x}$ and $ax\log(b/x)$ are both concave), we get

$$\begin{aligned}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})] &\le C\sqrt{\mathbb{E}[\hat\sigma_n^2(\mathcal{G})nv\log(2AU/\hat\sigma_n(\mathcal{G}))]}\\ &\le C\sqrt{\mathbb{E}[\hat\sigma_n^2(\mathcal{G})]nv\log((2AU)^2/\mathbb{E}[\hat\sigma_n^2(\mathcal{G})])/2}\end{aligned}$$

From Lemma 31, we obtain

$$\mathbb{E}[\hat\sigma_n^2(\mathcal{G})] \le \sigma^2 + 4Un^{-1}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})].$$

Now remark that $x\log(b/x)$ is non-decreasing for $x \le b/e$. This is always satisfied for $x = \sigma^2 + 4Un^{-1}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})]$ because this quantity is smaller than $5U^2$ which itself is smaller than $4eU^2$ that is smaller than $b/e = (2AU)^2/e$. We therefore obtain

$$\begin{aligned}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})] &\le C\sqrt{nv(\sigma^2 + 4Un^{-1}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})])\log((2AU)^2/(\sigma^2 + 4Un^{-1}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})]))]/2}\\ &\le C\sqrt{nv(\sigma^2 + 4Un^{-1}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})])\log((2AU)^2/\sigma^2)/2}\end{aligned}$$

It follows that

$$\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})]^2 \leq C^2 nv(\sigma^2 + 4Un^{-1}\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})])\log(2AU/\sigma) = b\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})] + c$$

with $b = C^2 4Uv\log(2AU/\sigma)$ and $c = C^2 vn\sigma^2\log(2AU/\sigma)$. Using Lemma 61, we get $\mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G})] \leq b + \sqrt{c}$ and so comes the first result. The bound for $\mathbb{E}[\mathrm{Rad}_n(\mathcal{G})]$ is obtained noting that $\mathbb{E}[\mathrm{Rad}_n(\mathcal{G})] = \mathbb{E}[\mathrm{Rad}_{n,s}(\mathcal{G} \cup -\mathcal{G})]$ and that the class $\mathcal{G} \cup -\mathcal{G}$ has a covering number twice the size as the one of $\mathcal{G}$. It is then of VC type with parameter v and $2^{1/v}A \leq 2A$. □

Combining Theorem 35 and Bousquet's inequality, Theorem 17, we obtain the following.

Theorem 36. *Let $(Z, Z_1, \ldots, Z_n)$ be an independent and identically distributed collection of random variables in $(S, \mathcal{S})$. Let $\mathcal{G}$ be a VC class of functions with parameters $v \geq 1$, $A \geq 1$ and uniform envelope $U \geq \sup_{g\in\mathcal{G}, x\in S} |g(x)|$. Let σ be such that $\sigma^2 \geq \sup_{g\in\mathcal{G}} Pg^2$ and $\sigma \leq U$. For any $n \geq 1$ and $\delta \in (0,1)$, it holds, with probability at least $1-\delta$,*

$$\sup_{g\in\mathcal{G}} \left| \sum_{i=1}^n \{g(Z_i) - P(g)\} \right| \leq \sqrt{32C^2 vn\sigma^2 \log(4AU/(\delta\sigma))} + 16C^2 Uv\log(4AU/(\delta\sigma))$$

with $C = 8$.

Proof. Recall that $Z_{\mathcal{G}} = \sup_{g\in\mathcal{G}} |\sum_{i=1}^n \{g(Z_i) - P(g)\}|$. Combining Theorem 17 and symmetrization inequality we get

$$Z_{\mathcal{G}} \leq 4\mathbb{E}[\mathrm{Rad}_n(\mathcal{G})] + \sqrt{2n\sigma^2\log(1/\delta)} + 8U\log(1/\delta)/3 \tag{1}$$

Relying on Theorem 35 to bound $\mathbb{E}[\mathrm{Rad}_n(\mathcal{G})]$, we get

$$Z_{\mathcal{G}} \leq 4C\sqrt{vn\sigma^2\log(4AU/\sigma)} + 16C^2 Uv\log(4AU/\sigma) + \sqrt{2n\sigma^2\log(1/\delta)} + 8U\log(1/\delta)/3.$$

For the two square-root terms, writing $p = 4C\sqrt{vn\sigma^2\log(4AU/\sigma)}$ and $q = \sqrt{2n\sigma^2\log(1/\delta)}$, we use $p + q \leq \sqrt{2(p^2+q^2)}$, where $p^2 = 16C^2 vn\sigma^2\log(4AU/\sigma)$:

$$p + q \;\leq\; \sqrt{2\big(16C^2 vn\sigma^2\log(4AU/\sigma) + 2n\sigma^2\log(1/\delta)\big)} \;=\; \sqrt{32C^2 vn\sigma^2\log(4AU/\sigma) + 4n\sigma^2\log(1/\delta)}.$$

Using $v \geq 1$ and $32C^2 \geq 4$ (as $C = 8$), $4n\sigma^2\log(1/\delta) \leq 32C^2 vn\sigma^2\log(1/\delta)$, so

$$p + q \;\leq\; \sqrt{32C^2 vn\sigma^2\big(\log(4AU/\sigma) + \log(1/\delta)\big)} \;=\; \sqrt{32C^2 vn\sigma^2\log(4AU/(\delta\sigma))}.$$

For the two remaining terms, using $v \geq 1$,

$$16C^2 Uv\log(4AU/\sigma) + \frac{8U}{3}\log(1/\delta) \;\leq\; 16C^2 Uv\log(4AU/(\delta\sigma))$$

since $16C^2 \geq 8/3$. Combining,

$$Z_{\mathcal{G}} \;\leq\; \sqrt{32C^2 vn\sigma^2\log(4AU/(\delta\sigma))} + 16C^2 Uv\log(4AU/(\delta\sigma)). \quad \square$$

Theorem 37. *In Theorem 36, if*

$$n\left(\frac{\sigma}{U}\right)^2 \geq 8C^2 v\log(4AU/(\delta\sigma))$$

it holds, with probability at least $1-\delta$,

$$\sup_{g\in\mathcal{G}} \left| \sum_{i=1}^n \{g(Z_i) - P(g)\} \right| \leq 64\sqrt{2}\sqrt{vn\sigma^2\log(4AU/(\delta\sigma))}$$

Proof. Write $Z_{\mathcal{G}} \leq S + L$ with $S = \sqrt{32C^2vn\sigma^2 \log(4AU/(\delta\sigma))}$ and $L = 16C^2Uv \log(4AU/(\delta\sigma))$ from Theorem 36. The stated condition is equivalent, by squaring, to $L \leq S$: indeed $L \leq S \iff L^2 \leq S^2 = 32C^2vn\sigma^2 \log(\cdot) \iff 256C^4U^2v^2 \log(\cdot)^2 \leq 32C^2vn\sigma^2 \log(\cdot) \iff 8C^2U^2v \log(\cdot) \leq n\sigma^2$. Under this condition, $Z_{\mathcal{G}} \leq S+L \leq 2S = 2\sqrt{32}\, C\sqrt{vn\sigma^2 \log(4AU/(\delta\sigma))} = 8\sqrt{2}\, C\sqrt{vn\sigma^2 \log(4AU/(\delta\sigma))}$. With $C = 8$, $8\sqrt{2}\, C = 64\sqrt{2}$. □

6.3 Examples

6.3.1 Indicator of cells

Due to its use in the definition of cumulative distribution function, one standard example the class of indicators of the cells $(-\infty, t]$, when $t \in \mathbb{R}$. There is actually two ways to deal with this class. The first one, which is presented in this paragraph, is using bracketing numbers (which is another notion of complexity). The second one is based on the shattering coefficient which will be presented next. Let us introduce bracketing numbers.

Definition 38 (bracketing numbers)**.** *Let $\underline{f}$ and $\overline{f}$ be two functions in $L_2(Q)$. The set $[\underline{f}, \overline{f}]$, called a bracket, denotes the set of all functions g in $L_2(Q)$ such that $\underline{f} \leq g \leq \overline{f}$. A bracket $[\underline{f}, \overline{f}]$ such that $\|\underline{f} - \overline{f}\|_{L_2(Q)} \leq \epsilon$ is called an ϵ-bracket. Given $\mathcal{G} \subset L_2(Q)$, the ϵ-bracketing number, denoted $\mathcal{N}_{[]}(\mathcal{G}, L_2(Q), \epsilon)$, is defined as the smallest number of brackets of radius $\epsilon > 0$ needed to cover $\mathcal{G}$.*

The next lemma consider the bracketing numbers associated to the class of cells in $\mathbb{R}$.

$$\mathcal{I} = \{z \mapsto \mathbb{1}_{z \leq t} \, : \, t \in \mathbb{R}\}$$

Lemma 39. *For all $\epsilon \in (0, 1/2)$ and any probability measure Q, we have that $N_{[]}(\mathcal{I}, L_1(Q), 2\epsilon) \leq 1/\epsilon$*

Proof. Let F be the cumulative distribution function of measure F. Define $F^-(u) = \inf\{t \in \mathbb{R} \, : \, F(t) \geq u\}$ and recall that $F_- \circ F^- \leq id \leq F \circ F^-$. Let $\epsilon \in (0, 1)$. Let $u_k = \epsilon k$, $k = 0, 1 \ldots, N-1$ with N being smallest integer such that $N\epsilon \geq 1$. Consequently, $N = \lceil 1/\epsilon \rceil$. Define $t_i = F^-(u_i), i = 0, \ldots, N-1$ and set $t_N = \infty$. The associated brackets, for any $i = 1, \ldots, N$

$$[\mathbb{1}_{(-\infty, t_{i-1}]} \, , \, \mathbb{1}_{(-\infty, t_i)}].$$

One has, for any i,

$$Q(\mathbb{1}_{(-\infty, t_i)} - \mathbb{1}_{(-\infty, t_{i-1}]}) = F_-(t_i) - F(t_{i-1}) \leq \epsilon$$

The number of brackets is $N = \lceil 1/\epsilon \rceil$. Then for all $\epsilon \in (0, 1)$, $N_{[]}(\mathcal{I}, L_1(Q), \epsilon) \leq 1/\epsilon + 1 \leq 2/\epsilon$, for all probability measure Q on $\mathbb{R}$. □

Interestingly, it holds that (this is one of the exercises)

$$\mathcal{N}(\mathcal{G}, L_p(Q), \epsilon) \leq \mathcal{N}_{[]}(\mathcal{G}, L_p(Q), 2\epsilon).$$

As a consequence, we deduce the following statement, that the class of cells is VC with parameter (Van Der Vaart and Wellner, 1996, Example 2.5.4).

Lemma 40. *For any probability measure Q, we have for all $\epsilon \in (0, 1/2)$, $\mathcal{N}(\mathcal{I}, L_1(Q), \epsilon) \leq 1/\epsilon$ and, for all $\epsilon \in (0, 1/\sqrt{2})$, $\mathcal{N}(\mathcal{I}, L_2(Q), \epsilon) \leq 1/\epsilon^2$.*

The previous might be refined in $\lceil 1/(2\epsilon^2) \rceil$. Also it might be extended to the multidimensional case constructing multidimensional brackets using product of 1-dimensional bracket. We obtain that $\mathcal{N}_{[]}(\mathcal{I}, L_2(Q), \epsilon)$ is of order $(1/\epsilon^2)^d$ when the dimension is $d \geq 1$. A similar bound shall be given relying on shattering numbers introduced next.

6.3.2 Parametric classes

The following result is well-known and proved for instance in Zhang (2023) (Theorem 5.3). It relies on Packing numbers, a concept similar to covering number, which we do not introduce here.

Lemma 41. *For any norm $\|\cdot\|$ on $\mathbb{R}^d$. We have, for all $\epsilon \in (0, r]$,*

$$\mathcal{N}(B(0,r), \|\cdot\|, \epsilon) \leq (1 + 2r/\epsilon)^d$$

As a consequence, for all $\epsilon \in (0, 1]$

$$\mathcal{N}(B(0,r), \|\cdot\|, \epsilon r) \leq (3/\epsilon)^d$$

Now now consider the parametric class

$$\mathcal{F}_\Theta := \{f_\theta - f_{\theta_0} \,:\, \theta \in \Theta\}$$

Lemma 42. *Suppose that Θ is bounded and that $\theta \mapsto f_\theta(x)$ is L-Lipschitz where L depends on x. Then for all $\epsilon \in (0, 1]$,*

$$\mathcal{N}(\mathcal{F}_\Theta, L_2(Q), \epsilon r_\Theta \|L\|_{L_2(Q)}) \leq (3/\epsilon)^d$$

where r_Θ is the radius of the smallest ball with center θ_0 that contains Θ.

In the above, note that $r_\Theta L$ is a proper envelop for the class of interest.

6.3.3 Indicator of balls and hyperplanes

The Vapnik-Chervonenkis dimension turns out to be related to the VC property. Roughly speaking, indicator functions of sets in $\mathcal{A}$ with finite Vapnik-Chervonenkis dimension are of VC type with in particular $v = 2vc(\mathcal{A})$. This is stated in the next Lemma which is due to Haussler (1995); see also Lemma 13.6 in Boucheron et al. (2013).

Lemma 43 (Haussler)**.** *For any probability measure Q, we have, for all $\epsilon \in (0, 1)$,*

$$\mathcal{N}(\mathbb{1}_\mathcal{A}, L_1(Q), \epsilon) \leq (e^4/\epsilon)^{vc(\mathcal{A})}$$
$$\mathcal{N}(\mathbb{1}_\mathcal{A}, L_2(Q), \epsilon) \leq (e^2/\epsilon)^{2vc(\mathcal{A})}$$

and the centres of the covering are elements of $\mathbb{1}_\mathcal{A}$.

The previous lemma is actually really useful as many popular classes of sets have a known finite Vapnik dimension as explained in Section 4. We have, for cells $\mathcal{C} = \{\prod_{k=1}^d (-\infty, b_k] \,:\, (b_1, \ldots, b_d) \in \mathbb{R}^d\}$ and rectangles $\mathcal{R} = \{\prod_{k=1}^d (-a_k, b_k]\}$, we have

$$\mathcal{N}(\mathbb{1}_\mathcal{C}, L_2(Q), \epsilon) \leq (e^2/\epsilon)^{2d} \qquad \mathcal{N}(\mathbb{1}_\mathcal{R}, L_2(Q), \epsilon) \leq (e^2/\epsilon)^{4d}.$$

Similarly, the class of balls defined as, $\mathcal{B} = \{B(x,r) : x \in \mathbb{R}^d, r > 0\}$ has Vapnik dimension equal to $d+1$. As a consequence,

$$\mathcal{N}(\mathbb{1}_{\mathcal{B}}, L_2(Q), \epsilon) \leq (e^2/\epsilon)^{2(d+1)}.$$

We also have the following result about collections of hyperplanes $\mathcal{H} = \{\langle w, x\rangle > 0 : w \in \mathbb{R}^d\}$, that is

$$\mathcal{N}(\mathbb{1}_{\mathcal{H}}, L_2(Q), \epsilon) \leq (e^2/\epsilon)^{2d}.$$

6.3.4 Kernel functions

In the previous section, the Vapnik dimension, a property related to subsets, has been employed to deal with particular class of functions defined as indicators of sets. Here we consider general class of functions (not necessarily indicators of sets) and we show that the VC property can still be useful. The relevant property for functions is related to the Vapnik dimension of their associated subgraph.

Definition 44. *A function class $\mathcal{F}$ is called VC-subgraph whenever the class $\mathcal{C}$ of all its subgraphs $\mathcal{C}_f = \{(x,t) : t < f(x)\}$ has a finite Vapnik dimension $vc(\mathcal{C})$.*

Lemma 45. *Let $\mathcal{F}$ be a VC subgraph class, we have*

$$\mathcal{N}(\mathcal{F}, L_2(Q), \epsilon\|F\|_{L_2(Q)}) \leq (2e^2/\epsilon)^{2vc(\mathcal{C})}$$

Proof. The proof is adapted from Van Der Vaart and Wellner (1996) Theorem 2.6.7. We have

$$Q(|f-g|) = Q((f-g)1_f > g) + Q((g-f)1_f < g) = Q \times \lambda(1_{g\leq t<f} + 1_{f\leq t<g})$$

But the function $1_{g\leq t<f} + 1_{f\leq t<g}$ is equal to $1_{\mathcal{C}_f \Delta \mathcal{C}_g}$. Let $S = \mathcal{C}_F \Delta \mathcal{C}_{-F}$. One may check that $P(A) = Q \times \lambda(A \cap S)/2QF$ is a probability measure on the set $S = \{-F(x) \leq t < F(x)\}$. From Lemma 43, there exists $\mathbb{1}_{A_k}$, $k = 1 \dots N_\epsilon$, where A_k is a subgraph, a ϵ cover of $\mathcal{C}$. Let f_k be the associated function. We have

$$Q(|f-f_k|) = P(\mathcal{C}_f \Delta \mathcal{C}_k) 2QF \leq 2QF\epsilon$$

As a consequence,

$$\mathcal{N}(\mathcal{F}, L_1(Q), \epsilon 2QF) \leq \mathcal{N}(\mathbb{1}_{\mathcal{C}}, L_1(P), \epsilon) \leq (e^4/\epsilon)^{vc(\mathcal{A})}.$$

We have a result for $L_1(Q)$. To obtain it for $L_2(Q)$, note that

$$Q(f^2) = Q(2F)\tilde{Q}(f)$$

where $\tilde{Q} = Q/Q(2F)$ is another probability measure. Take f_k a $L_1(\tilde{Q})$ cover of size $\epsilon^2 Q(2F)$ of $\mathcal{F}$. As a consequence, $Q(f-f_k)^2 = \tilde{Q}(|f-f_k|)Q(2F) \leq \epsilon^2 Q(2F)^2 \leq \epsilon^2 Q((2F^2)$. Therefore,

$$\mathcal{N}(\mathcal{F}, L_2(Q), \epsilon\|2F\|_{L_2(Q)}) \leq \mathcal{N}(\mathcal{F}, L_1(\tilde{Q}), \epsilon^2 Q(2F)) \leq (e^4/\epsilon^2)^{vc(\mathcal{A})}.$$

□

We conclude this section by another result which will be useful for kernel estimate. A more general result is in Giné and Nickl (2021) Proposition 3.6.12. This type of ideas were developed in Nolan and Pollard (1987).

Lemma 46. *Suppose that $K : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a decreasing function. Then the subgraphs of*

$$\mathcal{K} = \{x \mapsto K(\|h^{-1}(x-s)\|) \, : \, h > 0, s \in \mathbb{R}^d\}$$

have a Vapnik dimension $d+3$. As a consequence

$$\mathcal{N}(\mathcal{K}, L_2(Q), \epsilon \|K\|_{L_2(Q)}) \leq (2e^2/\epsilon)^{2(d+3)}$$

Proof. The proof use that subgraphs of the previous might be written as

$$\|h^{-1}(x-s)\|^2 - K^{-1}(t)^2 > 0$$

but $\|h^{-1}(x-s)\|^2 = h^{-2}\|x\|^2 - 2h^{-2}\langle x, s\rangle - K^{-1}(t)^2 + h^{-1}\|s\|^2$ is included the space generated by $(x,t) \mapsto (\|x\|^2, x, K^{-1}(t)^2, 1)$. The dimension is $d+3$ based on Lemma 24. The conclusion follows from Theorem 45.

□

When h is considered fixed, the approach can be simplified in the real case as indicated in Giné and Nickl (2009).

Lemma 47. *Suppose that $K : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a decreasing function. Then the subgraphs of*

$$\mathcal{K} = \{x \mapsto K(x-s) \, : \, s \in \mathbb{R}\}$$

have a Vapnik dimension 2.

The previous results will be useful to obtain a uniform bound with respect to x for the KDE estimate.

6.4 Preservation properties

We next call an envelope for $\mathcal{G}$ any function $G : S \mapsto \mathbb{R}$ that satisfies $|g(x)| \leq G(x)$ for all $x \in S$ and $g \in \mathcal{G}$. We first describe some basic preservation property related to VC classes.

Lemma 48. *Suppose that $\mathcal{F}$ and $\mathcal{G}$ are both VC. Then $-\mathcal{F}$, $\mathcal{F} \cup \mathcal{G}$, $\mathcal{F} + \mathcal{G}$, $\mathcal{F}\mathcal{G}$ are VC*

Lemma 49. *Suppose that $\mathcal{F}$ (resp. $\mathcal{G}$) defined on $(S, \mathcal{S})$ is of VC-type with envelope U and parameter (v, A) and let $E \in \mathcal{S}$. The following holds:*

1. *$\{fI_E \, : \, f \in \mathcal{F}\}$ is of VC-type with envelope U and parameter (v, A),*
2. *$\mathcal{F} - \mathcal{G} = \{f - g \, : \, f \in \mathcal{F}, g \in \mathcal{G}\}$ is of VC-type with envelope $2U$ and parameter $(2v, 2A)$,*
3. *$\{f - P(f|E) \, : \, f \in \mathcal{F}\}$ is of VC-type with envelope $2U$ and parameter $(2v, A)$,*
4. *$\{qf + (1-q)g \, : \, f \in \mathcal{F}, g \in \mathcal{G}, q \in [0,1]\}$ is of VC-type with envelope U and parameter $(2v+1, 3A)$.*

Proof. Let Q be a probability measure on $(S, \mathcal{S})$. Let $(f_k)_{k=1,\dots,K}$ be the center of an ϵU-covering of $(\mathcal{F}, Q)$. The first statement follow from the fact that $\|f1_E - f_k 1_E\|_{L_2(Q)} \leq \|f - f_k\|_{L_2(Q)}$. For the second statement consider $U\epsilon$-covers $(f_k, k \leq K)$ and $(g_j, j \leq J)$ respectively of $\mathcal{F}$ and $\mathcal{G}$. Then the triangle inequality shows that $(f_k - g_j), k \leq K, j \leq J$ forms a $2U\epsilon$-cover of $\mathcal{F} + \mathcal{G}$ and the result follows. Now let $(\tilde{f}_k)_{k=1,\dots,K}$ be the center of an ϵU-covering of $(\mathcal{F}, P_E)$ with $P_E(\cdot) = P(\cdot|E)$. Consider the covering induces by the centers $(f_k - P_E(\tilde{f}_j))_{1\leq k,j\leq K}$ made of K^2 elements. Suppose that $f \in \mathcal{F}$. Then there is k and j such that

$$\begin{aligned}\|(f - P_E(f)) - (f_k - P_E(\tilde{f}_j)\|_{L_2(Q)} &\leq \|f - f_k\|_{L_2(Q)} + P_E(f - \tilde{f}_j) \\ &\leq \|f - f_k\|_{L_2(Q)} + \|f - \tilde{f}_j\|_{L_2(P)} \\ &\leq 2U\epsilon.\end{aligned}$$

Hence we have found a $2U\epsilon$-covering of size K^2 which by assumption is smaller than $(A/\epsilon)^{2v}$. This implies the third statement of the lemma. For the last statement, let $\mathcal{H} = \{qf + (1-q)g : f \in \mathcal{F},\, g \in \mathcal{G},\, q \in [0,1]\}$. Let $(f_k)_{k=1,\dots,K}$ (resp. g_ℓ, $(g_\ell)_{\ell=1,\dots,L}$) be the center of an ϵU-covering of $(\mathcal{F}, Q)$ (resp. $(\mathcal{G}, Q)$). Let q_i, $i = 1, \dots, \lfloor 1/\epsilon \rfloor$ be an ϵ-covering of $[0,1]$. Let $h = qf + (1-q)g$ be such that $q \in [0,1]$, $f \in \mathcal{F}$ and $g \in \mathcal{G}$. There is f_k, g_ℓ, q_i such that

$$\begin{aligned}&\|qf + (1-q)g - (q_i f_k + (1-q_i)g_\ell)\|_{L_2(Q)} \\ &\leq \|q(f - f_k) + (1-q)(g - g_\ell)\|_{L_2(Q)} + \|(q - q_i)f_k + (q_i - q)g_\ell\|_{L_2(Q)} \\ &\leq qU\epsilon + (1-q)U\epsilon + \epsilon U + \epsilon U = 3\epsilon U.\end{aligned}$$

Hence the element $(q_i f_k + (1-q_i)g_\ell)$ form an $3\epsilon U$-covering in the space $L_2(Q)$. There are $KL\lceil 1/\epsilon \rceil \leq KL/\epsilon$ such elements. As a consequence, since $\mathcal{F}$ and $\mathcal{G}$ are are of VC-type, it follows that

$$\mathcal{N}(\mathcal{H}, L_2(Q), 3\epsilon U) \leq (A/\epsilon)^{2v}(1/\epsilon) \leq (A/\epsilon)^{2v+1}$$

which implies the stated result.

□

Define the set

$$Star(\mathcal{G}) = \{\alpha g : \alpha \in [0,1],\, g \in \mathcal{G}\}.$$

Proposition 50. *If $\mathcal{G}$ is of VC type with envelop $\|G\|$ and parameters (v, A) then $Star(\mathcal{G})$ is of VC type with envelop $\|G\|$ and parameter $(v+1, A)$.*

Proof. Let Q be a probability measure on $(S, \mathcal{S})$ and $\epsilon \in (0,1)$. Let $g_1, \dots, g_J$ be a $\|G\|_{L_2(Q)}\epsilon/2$-cover of $\mathcal{G}$. By definition $J \leq (2A/\epsilon)^v$. Let $K = \lceil 1/\epsilon \rceil$. The following set made of K balls $[0,\epsilon], [\epsilon, 2\epsilon], \dots, [(K-1)\epsilon, K\epsilon]$ is an $\epsilon/2$-cover of the space $[0,1]$. The centres of these balls are $\alpha_k = (2k-1)\epsilon/2$, $k = 1, \dots, K$. Consider the collection of functions $\{\alpha_k g_j,\, k = 1, \dots, K,\, j = 1, \dots, J\}$. Let $\alpha g \in Star(\mathcal{G})$. Then there is $\alpha_k g_j$ such that

$$\|\alpha g - \alpha_k g_j\|_{L_2(Q)} \leq \|(\alpha - \alpha_k)g\|_{L_2(Q)} + \|\alpha_k(g - g_j)\|_{L_2(Q)} \leq \epsilon\|G\|_{L_2(Q)}.$$

We have then found an $\epsilon\|G\|_{L_2(Q)}$-net of $(Star(\mathcal{G}), L_2(Q))$. The number of elements in the net is $(2A/\epsilon)^v K \leq (2A/\epsilon)^v(2/\epsilon) \leq (2A/\epsilon)^{v+1}$.

□

6.5 Exercises

Exercise 20. *1. Show that*

$$\mathcal{N}(\mathcal{G}, L_2(Q), \epsilon) \leq \mathcal{N}_{[]}(\mathcal{G}, L_2(Q), 2\epsilon).$$

2. Suppose that $\mathcal{F}$ and $\mathcal{G}$ both are valued in $[0,1]$. Show that

$$\mathcal{N}_{[]}(\mathcal{G}\mathcal{F}, L_2(Q), 2\epsilon) \leq \mathcal{N}_{[]}(\mathcal{F}, L_2(Q), \epsilon)\mathcal{N}_{[]}(\mathcal{G}, L_2(Q), \epsilon)$$

3. If $\mathcal{I}$ is the class of cells in dimension 2. Show that

$$\mathcal{N}_{[]}(\mathcal{I}, L_2(Q), 2\epsilon) \leq (2/\epsilon^2)^2$$

7 Local averaging rules

Local averaging is a common statistical approach which consists in "averaging" data restricted to some (often small) region of the space. This is contrasting with standard approach that would use the full data to estimate some unknown quantities depending on the underlying probability measure. Local averaging around a given point x is often useful in building estimator of "local" quantities that depends on the conditional probability measure given x, μ_x, instead of the common underlying measure μ. For instance, if one which to estimate the regression function $E[Y|X=x] = \int y \mathrm{d}\mu_x(y)$, then one could use the points among the initial sample that are nearby x by averaging them. Another example different from regression is density estimation as introduce in the next section.

7.1 Density estimation by kernel smoothing

In this section, we provide a uniform deviation result on the kernel estimate of the density $\hat{f}_n$ introduced in the first section. Results of this type can be found in Giné and Guillou (2001); Giné and Guillou (2002); Giné and Nickl (2009). We consider the following standard framework for density estimation.

(KDE) Let $(X, X_1, \ldots, X_n)$ be a sequence of independent and identically distributed random elements with common distribution P_X on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. We suppose that P_X has a density f with respect to Lebesgue measure, i.e., for all $B \in \mathcal{B}(\mathbb{R}^d)$, $P_X(B) = \int_B f \mathrm{d}\lambda$.

We wish to estimate f uniformly in $x \in \mathbb{R}^d$ using KDE $\hat{f}_n$ which has been shown to be a good point-wise estimator of f. Relying on the same decomposition as before, we have

$$\hat{f}_n(x) - f(x) = \hat{f}_n(x) - f_h(x) + f_h(x) - f(x).$$

To take care of the bias term, we consider the following regularity condition.

(Reg2) The function f is L smooth; that is, f is differentiable and $\nabla_x f(x)$ is L-Lipschitz

$$\|\nabla_x f(x) - \nabla_x f(y)\| \leq L\|x - y\|$$

The bias term has already been considered in a previous exercise but under a Lipschitz condition. Here we hope that the L-smoothness (which is stronger) will imply better convergence rate.

Theorem 51. *Suppose that (KDE) and (Reg2) are fulfilled. If*

$$n \geq nh^d V_d U_f \geq 16C^2(d+1)\log(4e^2/(\delta h^d V_d U_f))$$

then with probability $1-\delta$,

$$\sup_{x\in\mathbb{R}^d} |\hat{f}_n(x) - f(x)| \leq 2\sqrt{\frac{16C^2(d+1)U_f \log(4e^2 n/\delta)}{nh^d V_d}} + \frac{1}{2}h^2 L$$

Proof. Lemma 60 is useful to obtain appropriate rate of convergence for the bias term. Because $\int_{B(0,1)} u du = 0$, we have that

$$f_h(x) - f(x) = V_d^{-1} \int_{B(0,1)} f(x - hu) - f(x) dx = V_d^{-1} \int_{B(0,1)} f(x - hu) - f(x) - hu^T \nabla_x f(x) dx$$

Therefore

$$|f_h(x) - f(x)| \leq V_d^{-1} \int_{B(0,1)} |f(x - hu) - f(x) - hu^T \nabla_x f(x)| dx \leq \frac{1}{2} V_d^{-1} h^2 L \int_{B(0,1)} \|u\|^2 dx$$

which gives

$$\sup_{x \in \mathbb{R}^d} |f_h(x) - f(x)| \leq \frac{1}{2} h^2 L$$

Now let us consider the variance part. We have

$$\hat{f}_n(x) - f_h(x) = n^{-1} h^{-d} V_d^{-1} \sum_{i=1}^{n} \{g_{x,h}(X_i) - P(g_{x,h})\}$$

with $g_{x,h}(X) = \mathbb{1}_{B(x,h)}(X)$. The class $\{g_{x,h} \;\; x \in \mathbb{R}^d, h > 0\}$ is VC because it is included in the set of indicator of balls. The parameter are $(e^2, 2(d+1))$. The class $\{g_{x,h} \;\; x \in \mathbb{R}^d\}$ is also VC because it is included in the previous class. Note that the VC parameter do not depend on h. Applying our main result, Theorem 37, we need to choose the variance value σ^2. We have

$$\sup_x P(g_{x,h}^2) = \sup_x P(g_{x,h}) \leq h^d V_d U_f = \sigma^2$$

so we need to assume that $h^d V_d U_f \leq 1$. If in addition

$$nh^d V_d U_f \geq 16 C^2 (d+1) \log(4e^2/(\delta h^d V_d U_f))$$

we obtain that, with probability $1 - \delta$,

$$\sup_{x \in \mathbb{R}^d} |\hat{f}_n(x) - f_h(x)| \leq 2 \sqrt{\frac{16 C^2 (d+1) U_f \log(4e^2/(\delta h^d V_d U_f))}{nh^d V_d}}$$

Note that since $h^d V_d U_f \leq 1$ it holds that $\log(4e^2/(\delta h^d V_d U_f)) \geq 1$ and in particular $nh^d V_d U_f \geq 1$. Hence using this lower bound on h in previous upper bound we obtain the stated result.

□

To emphasize the strength of previous result, one may use it when $\delta = n^{-2}$ to obtain that whenever

$$nh^d / \log(n) \to \infty, \qquad h \to 0$$

we have, with probability 1, as $n \to \infty$,

$$\sup_{x \in \mathbb{R}^d} |\hat{f}_n(x) - f_h(x)| = O\left(\sqrt{\frac{\log(n)}{nh^d}} + h^2 \right).$$

Optimizing the above with respect to h gives $h = (\log(n)/n)^{1/(d+4)}$ which is the optimal choice of h for twice differentiable functions. If the density would have been only Lipschitz then we would

have obtained $h = (\log(n)/n)^{1/(d+2)}$. Each regularity scenario leads to different rates, $n^{-2/(d+4)}$ and $n^{-1/(d+2)}$, which are the optimal ones in the minimax sense Tsybakov (2008). More generally, when f is $\mathcal{C}^s$, the optimal rate of convergence is $(\log(n)/n)^{s/(d+2s)}$. The latter facts emphasizes the difficulty to select the bandwidth in practice as we do not know in general the regularity of the function. For choosing the bandwidth, adaptive methods aim to achieve the optimal choice without knowing the smoothness of the function. Such methods rely on bandwidth selecting procedure for which $\hat{h}_n$ depends on the sample. The results proposed before might be useful in that case because of the VC property of balls we have to show that $\hat{h}$ lies in the appropriate range so that the above bound goes to 0. For instance, as soon as the $\hat{h}_n$ is such that

$$n\hat{h}_n^d \to \infty \qquad \hat{h}_n \to 0$$

One might obtain the consistency of adaptive methods. This does not imply that they would be optimal. More involved results are that adaptive choices can be optimal with respect to the regularity of the associated function f. The previous estimator can be modified to estimate the regression function. Define

$$\hat{g}_n(x) = \frac{\sum_{i=1}^n Y_i \mathbb{1}_{B(x,h)}(X_i)}{\sum_{i=1}^n \mathbb{1}_{B(x,h)}(X_i)}$$

7.2 Nearest neighbor algorithm

Consider the following k-NN estimate $\mu_x(g) = \mathbb{E}[g(Y)|X = x]$ given by

$$\hat{\mu}_n(g) = k^{-1} \sum_{i \in N_{n,k}(x)} g(Y_i).$$

Some additional assumptions will be needed to control the size of the k-NN neighborhood.

(X1) There is $c > 0$ and $T > 0$ such that

$$\lambda(S_X \cap B(x, \tau)) \geq c\lambda(B(x, \tau)), \qquad \forall \tau \in (0, T], \forall x \in S_X,$$

where λ is the Lebesgue measure.

(X2) There is $0 < b_X \leq U_X < +\infty$ such that $b_X \leq f(x) \leq U_X$, for all $x \in S_X$.

It is useful to derive an upper bound on the k-NN k-NN radius. Define

$$\overline{\tau}_{n,k} = \left(\frac{2k}{nb_X c V_d}\right)^{1/d}.$$

The first above condition is interesting for small $\tau > 0$ and hence deals with the boundary of the set S_X while the second means that the measure P_X charges uniformly the set S_X (no region with no point).

The following Lemma controls the size of the k-NN balls uniformly over all $x \in S_X$. In this way, it extends the multiplicative Chernoff bound which focuses on a particular x. The proof is based on the second Vapnik inequality. We stress that a similar result (with less advantageous constants) could be obtained using Theorem 37 instead of Theorem 21.

Lemma 52. *Suppose that (H), (X1) and (X2) hold true. Then, for all $n \geq 1$, $\delta \in (0,1)$ and $1 \leq k \leq n$ such that $16d\log(12n/\delta) \leq k \leq T^d n b_X c V_d/2$, it holds, with probability at least $1-\delta$:*

$$\sup_{x \in S_X} \hat{\tau}_{n,k,x} \leq \overline{\tau}_{n,k}.$$

Proof. The proof is similar the proof of Lemma 8. The difference is that we use the second Vapnik inequality, instead of Theorem 1 so that the reasoning is uniform over x. We have that, for all $x \in S_X$,

$$P(B(x,\overline{\tau}_{n,k})) = \int_{S_X \cap B(x,\overline{\tau}_{n,k})} f_X(x)dx \geq bc\lambda(B(x,\overline{\tau}_{n,k})) = 2\frac{k}{n}.$$

Lemma 21 indicates that, with probability $1-\delta$, for all $x \in S_X$,

$$n\mathbb{P}_n(B(x,\overline{\tau}_{n,k})) \geq nP(B(x,\overline{\tau}_{n,k})) - \sqrt{nP(B(x,\overline{\tau}_{n,k}))8d\log(12n/\delta)}$$

Noting that $x \mapsto x - \sqrt{x\ell}$ is increasing whenever $x \geq \ell/4$, we obtain, since $2k \geq (16/4)kd\log(12n/\delta)$,

$$n\mathbb{P}_n(B(x,\overline{\tau}_{n,k})) \geq 2k - \sqrt{16kd\log(12n/\delta)}.$$

The above quantity is larger than k whenever $k \geq \sqrt{16kd\log(12n/\delta)}$, equivalently, $k \geq 16d\log(12n/\delta)$. Now it remains to use that, for each x, $\hat{\tau}_{n,k,x}$ is the smaller such τ so it needs to be smaller than $\overline{\tau}_{n,k}$. □

In the standard regression setup, $g(Y) = Y$, as in second chapter and under assumption (E). The variance is given by

$$v_n(x) = k^{-1}\sum_{i=1}^{n} \epsilon_i \mathbb{1}_{B(x,\hat{\tau}_{n,k,x})}(X_i)$$

Conditionally on $X_1,\ldots,X_n$, the previous quantity is sub-Gaussian with factor $\sigma^2 \sum_{i=1}^n \epsilon_i \mathbb{1}_{B(x,\hat{\tau}_{n,k,x})}(X_i) = \sigma^2 k$. Since $B(x,\hat{\tau}_{n,k,x})$ is included in the set of balls in $\mathbb{R}^d$. Conditionally on $X_1,\ldots,X_n$, there is at most $(n+1)^v$, $v = d+1$, elements in the sets $(\mathbb{1}_{B(x,\hat{\tau}_{n,k,x})}(X_i))_{i=1,\ldots,n}$. Therefore we can apply Vapnik type inequality, Theorem 20, to obtain that

$$\mathbb{P}(\sup_{x \in S_X} |v_n(x)| > t | X_1,\ldots,X_n) \leq 2(n+1)^v \exp(-t^2 k/(8\sigma^2))$$

Taking the expectation and changing variables gives with probability at least $1-\delta$,

$$\sup_{x \in S_X} |v_n(x)| \leq \sqrt{\frac{8\sigma^2 \log(2(n+1)^v)}{k}}.$$

We now state our main result, a non-asymptotic bound on the error associated to the nearest neighbor regression estimator. The proposed bound on the k-NN process should hold uniformly in $x \in S_X$ where the set S_X stands for the support of P_X.

Theorem 53. *Suppose that (H), (X1), (X2) are fulfilled and that $x \mapsto \mu_x(g)$ is L-Lipschitz. Let $\gamma = U_g U_f/b_f$. Then whenever*

$$n \geq 2\gamma k \geq 16C^2(d+1)\log(8e^2 n/\delta)$$

it holds with probability $1-\delta$,

$$|\hat{\mu}_x(g) - \mu_x(g)| \leq 2\sqrt{\frac{16C^2(d+1)2\gamma}{k}\log(8e^2 n/\delta)} + L\overline{\tau}_{n,k}$$

Proof. Note that the estimator of interest may be written as

$$\hat{\mu}_x(g) = k^{-1} \sum_{i=1}^{n} g(Y_i) \mathbb{1}_{B(x,\hat{\tau}_{n,k,x})}(X_i).$$

Write

$$\hat{\mu}_x(g) - \mu_x(g) = k^{-1} \sum_{i \in N_{n,k}(x)} (g(Y_i) - \mu_{X_i}(g)) + k^{-1} \sum_{i \in N_{n,k}(x)} (\mu_{X_i}(g) - \mu_x(g))$$

The second term in the left-hand side is smaller than

$$Lk^{-1} \sum_{i \in N_{n,k}(x)} \|X_i - x\| \leq L\hat{\tau}_{n,k,x}$$

Using Lemma 52, we have that, with probability $1 - \delta$,

$$\hat{\tau}_{n,k,x} \leq \overline{\tau}_{n,k}$$

For the first term in the left-hand side, we have that, with probability $1 - \delta$,

$$\sum_{i \in N_{n,k}(x)} (Y_i - g(X_i)) \leq \sup_{x \in S_X, \tau \leq \tau_{n,k}} \sum_{i=1}^{n} (g(Y_i) - \mu_{X_i}(g)) \mathbb{1}_{B(x,\tau)}(X_i)$$

The above is a centred empirical process to which we apply the Theorem 37 with

$$\mathbb{E}[(g(Y) - \mu_X(g))^2 \mathbb{1}_{B(x,\tau)}(X)] \leq \| \operatorname{var}(g(Y)|x)\|_\infty \tau^d V_d U_f \leq U_g^2 U_f \overline{\tau}_{n,k}^d V_d = \sigma^2$$

Whenever $U_g^2 U_f \overline{\tau}_{n,k}^d V_d \leq 1$ and

$$nU_g^2 U_f \overline{\tau}_{n,k}^d V_d \geq 16C^2(d+1) \log(4e^2/(\delta U_g^2 U_f \overline{\tau}_{n,k}^d V_d))$$

we find, with probability $1 - \delta$,

$$\sum_{i \in N_{n,k}(x)} (Y_i - g(X_i)) \leq 2\sqrt{16C^2(d+1)nU_g^2 U_f \overline{\tau}_{n,k}^d V_d \log(4e^2/(\delta U_g U_f \overline{\tau}_{n,k}^d V_d))}$$

Using that $U_g^2 U_f \overline{\tau}_{n,k}^d V_d \leq 1$ we get that $nU_g^2 U_f \overline{\tau}_{n,k}^d V_d \geq 1$ and therefore, with probability $1 - \delta$,

$$\log(4e^2/(\delta U_g^2 U_f \overline{\tau}_{n,k}^d V_d)) \leq \log(4e^2 n/\delta)).$$

Conclude by taking $\delta/2$ in the above so that the two previous event are together realized with probability $1 - \delta$. □

Similar to before we have the following corollary that whenever $k/n \to 0$ and $k/\log(n) \to \infty$, with probability 1,

$$|\hat{\mu}_x(g) - \mu_x(g)| = O\left(\sqrt{\frac{\log(n)}{k}} + \overline{\tau}_{n,k}\right)$$

Choice of k The optimal choice of k is therefore of order $n^{2/d+2}$.

The variance In low density region, the variance term is smaller than the variance term of Nadaraya-Watson. As a consequence, if a non-asymptotic bound would be obtained for Nadaraya-Watson, it would be interested to compare it with the one given before.

7.3 Choice of k by hold-out

The Hold-out procedure is based on choosing the optimal k as follows:

$$\hat{k}'_m \in \arg\min_{k=1,\dots,n} P'_m(Y - \hat{h}_k)^2$$

where P'_m is the empirical measure based on another sample $(X'_1, Y'_1), \dots, (X'_m, Y'_m))$ which is independent from the initial sample. The objective is that the performance $P(h - \hat{h}_{\hat{k}'_m})^2$ is similar to that of k^* given by $\min_k P(h - \hat{h}_k)^2$.

Theorem 54. *Suppose that Y is bounded by U. Then with probability 1,*

$$\mathbb{E}' P(h - \hat{h}_{\hat{k}'_m})^2 \leq P(h - \hat{h}_{k^*})^2 + \sqrt{\frac{8U^4 \log(n)}{m}}$$

Proof. By definition of the conditional expectation

$$P(h - \hat{h}_{\hat{k}'_m})^2 - P(h - \hat{h}_{k^*})^2 = P(Y - \hat{h}_{\hat{k}'_m})^2 - P(Y - \hat{h}_{k^*})^2$$

and, in addition, we have$k^* \in \arg\min_k P(Y - \hat{h}_k)^2$. Therefore, using that $P'_m(Y - \hat{h}_{\hat{k}'_m})^2 \leq P'_m(Y - \hat{h}_{k^*})^2$, we obtain

$$P(h - \hat{h}_{\hat{k}'_m})^2 - P(h - \hat{h}_{k^*})^2 \leq (P - P'_m)(Y - \hat{h}_{\hat{k}'_m})^2 - (P'_m - P)(Y - \hat{h}_{k^*})^2$$

Take the expectation with respect to P'_m-sample, we get

$$\mathbb{E}'[P(h - \hat{h}_{\hat{k}'_m})^2] - P(h - \hat{h}_{k^*})^2 \leq \mathbb{E}'[(P - P'_m)(Y - \hat{h}_{\hat{k}'_m})^2] \leq \mathbb{E}'[\sup_{k=1,\dots,n} (P - P'_m)(Y - \hat{h}_k)^2]$$

We can now use Lemma 27 to obtain that

$$\mathbb{E}'[\sup_k (P - P'_m)(Y - \hat{h}_k)^2] \leq \sqrt{\frac{2s^2 \log(n)}{m}}$$

where s^2 stands for the sub-Gaussian factor of $Y - \hat{h}_k$ that we now derive. Since Y is bounded by U we have that

$$|Y - \hat{h}_k| \leq 2U$$

which implies that $(h - \hat{h}_k)^2 \leq 4U^2$ is sub-Gaussian with factor $4U^4$. Hence we obtain the stated result.

□

Note that the previous result provides a slight improvement in the constant compared to Theorem 16.1 in Biau and Devroye (2015). This result is interesting because whenever m is large enough (m larger than $n^{4/(d+2)} \log(n)$), using that $P(h - \hat{h}_{k^*})^2 = O_P(n^{-2/d+2})$, we obtain that

$$\mathbb{E}' P(h - \hat{h}_{\hat{k}'_m})^2 = O_P(n^{-2/(d+2)})$$

which is optimal for Lipschitz function.

7.4 Tree construction

In this section, we consider partitioning estimate based on data driven and greedy construction of the partitioning. This type of construction is sequential and at each step, the partition is refined. Those estimates are refereed to as regression tree. Any tree construction provide a set of leaf $\mathcal{L} \subset S_X$ that forms a partition of S_X. The regression estimate is then given by, for each $x \in S_X$,

$$\frac{\sum_{i=1}^n Y_i \mathbb{1}_{L(x)}(X_i)}{\sum_{i=1}^n \mathbb{1}_{L(x)}(X_i)}$$

where $L(x)$ denotes the unique element of the partition that contain x.

A popular tree construction consists in the following: for each leaf L:

- choose (k, t), k is the variable to split (among the relevant candidate) and t is the threshold (among the relevant value) , such that the resulting MSE is minimized
- Check for the stopping criterion

To apply the tools developed in previous sections, we need to have a particular stopping rule. This is important to obtain in the end the property that a leaf has enough points to control the variance and is small enough diameter to allow the bias to be small. The stopping rule that we consider is that the L threshold should be selected among thresholds that give a leaf diameter d_k larger than ℓh. The parameter ℓ is selected by the user. As a consequence, direction cannot be selected whenever its size d_k (size of the leaf along direction k) is smaller than $2\ell h$. A tree fully grown according to this rule is such that for all k, $\ell h \leq d_k \leq 2\ell h$. One can stop growing the tree whenever $\max_k d_k(L) \leq hL$. Given the previous rules, any direction and any threshold can be selected at each step.

Theorem 55. *For Lipschitz function, the rate of convergence of regression tree is of order* $1/\sqrt{n(\ell h)^d} + Lh$.

8 Fast rates in empirical risk minimization

In this section, we consider the following empirical risk minimization problem

$$\hat{f}_n = \arg\min_{f\in\mathcal{F}} \sum_{i=1}^{n} \ell(Y_i, f(X_i)) = \arg\min_{f\in\mathcal{F}} P_n(\ell_f),$$

with $\ell_f(X_i, Y_i) = \ell(Y_i, f(X_i))$, and write $f^* := \arg\min_{f\in\mathcal{F}} P(\ell_f)$ for the population risk minimizer, assumed to exist and to lie in $\mathcal{F}$. Throughout, $\|f\|_\infty \le U$ for all $f \in \mathcal{F}$, and Y takes values in a set $\mathcal{Y}$. Two examples are to be kept in mind: the squared loss for regression,

$$\ell(y, f) = (y - f)^2,$$

as well as the logistic loss for classification ($y \in \{-1, 1\}$),

$$\ell(y, f) = \log\big(1 + \exp(-yf)\big).$$

Shifting the loss by a function of y alone does not change $\hat{f}_n$ nor f^*, so we may assume $\ell(y, 0) = 0$, as is the case for both examples above. Moreover, we ask $f \mapsto \ell(y, f)$ to be L-Lipschitz, uniformly in y, as formalized in the next assumption.

(LIP) For all $y \in \mathcal{Y}$, $\ell(y, 0) = 0$. There is $L > 0$ such that for all $y \in \mathcal{Y}$ and all $f, f' \in [-U, U]$,

$$|\ell(y, f) - \ell(y, f')| \le L|f - f'|.$$

The logistic loss satisfies (LIP) with $L = 1$ for any U (and any $y \in \{-1, 1\}$). The squared loss does *not* satisfy a Lipschitz bound over all of $\mathbb{R}^3$ (the slope $|2y - f - f'|$ is unbounded in y); it satisfies (LIP) with $L = 2(U + U_Y)$ if $\mathcal{Y} \subset [-U_Y, U_Y]$, which is why the domain is restricted above. Recall that

$$\mathrm{Rad}_{n,s}(\mathcal{F}) = \mathbb{E}_\eta \sup_{f\in\mathcal{F}} \sum_{i=1}^{n} \eta_i f(X_i),$$

where (η_i) are i.i.d. Rademacher variables. Recall also the contraction principle (see (Boucheron et al., 2005, Theorem 11.6)). Under (LIP), we have

$$\mathrm{Rad}_{n,s}(\ell_{\mathcal{F}}) \le L\mathrm{Rad}_{n,s}(\mathcal{F}).$$

The starting point is the standard rate in the ERM problem: with probability 1,

$$P(\ell_{\hat{f}_n}) - P(\ell_{f^*}) \le 2 \sup_{f\in\mathcal{F}} |(P_n - P)(\ell_f)|.$$

In light of the results given previously, if the class $\mathcal{F}$ is VC, this yields a rate of order $\sqrt{v\sigma^2 n^{-1} \log(1/\sigma^2)}$ up to constants. Next we consider improving this rate under the so-called Bernstein condition $P(g^2) \le B_0\, Pg$ for all g in the class of interest. We follow Bartlett et al. (2005) (see also (Massart and Nédélec, 2006)). Let us first recall Theorem 3.3 in (Bartlett et al., 2005).

Theorem 56. *Let $\mathcal{G}$ be a class of functions with range $[a,b]$. Suppose that $\mathrm{var}(g) \le T(g) \le B_0 Pg$ holds for all $g \in \mathcal{G}$, for some constant $B_0 > 0$. Then for all $K > 1$ and $\delta \in (0,1)$, with probability $1-\delta$:*

$$\forall g \in \mathcal{G}, \qquad Pg \le \left(\frac{K}{K-1}\right) P_n g + \left(\frac{c_1 K r^*}{B_0}\right) + (11(b-a) + c_2 B_0 K) n^{-1} \log\left(\frac{1}{\delta}\right)$$

where r^* is the fixed point of a sub-root function $\psi(r)$ such that for all $r \ge r^*$,

$$\psi(r) \ge \frac{B_0}{n} \mathbb{E}\mathrm{Rad}_{n,s}(\mathcal{G} : T(g) \le r).$$

We now apply this result to the ERM problem, with $\mathcal{G}$ made of the functions $g(x,y) = \ell_f(x,y) - \ell_{f^*}(x,y)$, $f \in \mathcal{F}$. We rely on the following convexity assumption.

(CONV) There is $B > 0$ such that for all $f \in \mathcal{F}$

$$P((f-f^*)^2) \le BP(\ell_f - \ell_{f^*}).$$

For the squared loss, (CONV) holds with $B = 1$ whenever $\mathcal{F}$ is convex (and with equality if $\mathbb{E}[Y \mid X] = f^*(X)$). For the logistic loss, it holds with $B = 4 + 2e^U + 2e^{-U}$ when f^* coincides with the pointwise minimizer of the conditional risk and $|f^*| \le U$.

When combined with (LIP), the above condition implies the Bernstein condition of Theorem 56, with constant $B_L := BL^2$ in place of B_0. That is,

$$P((\ell_f - \ell_{f^*})^2) \le L^2 P((f-f^*)^2) \le BL^2\, P(\ell_f - \ell_{f^*}).$$

We choose $T(\ell_f - \ell_{f^*}) = L^2 P((f-f^*)^2)$ and apply Theorem 56 *with* $B_0 = B_L$. This gives the following statement.

Theorem 57. *Under (CONV) and (LIP), with probability $1-\delta$,*

$$P(\ell_{\hat{f}} - \ell_{f^*}) \le \frac{2c_1 r^*}{B_L} + (44LU + 2c_2 B_L)\, n^{-1} \log\left(\frac{1}{\delta}\right),$$

where r^* is the fixed point of a sub-root function $\psi(r)$ such that for all $r \ge r^*$,

$$\psi(r) \ge \frac{B_L}{n} \mathbb{E}\mathrm{Rad}_{n,s}\big(\ell_{\mathcal{F}} - \ell_{f^*} : L^2 P((f-f^*)^2) \le r\big).$$

Proof. Apply Theorem 56 with $\mathcal{G} = \{\ell_f - \ell_{f^*} : f \in \mathcal{F}\}$, $B_0 = B_L$ and $K = 2$, to $\ell_{\hat{f}} - \ell_{f^*}$. Since $\hat{f}$ minimizes $P_n(\ell_f)$ over $\mathcal{F} \ni f^*$, we have $P_n(\ell_{\hat{f}} - \ell_{f^*}) \le 0$, so the first term on the right-hand side of the theorem is non-positive. For the range: $\|f\|_\infty, \|f^*\|_\infty \le U$ gives $|f - f^*| \le 2U$, and by (LIP), $|\ell_f - \ell_{f^*}| \le L|f - f^*| \le 2LU$. Hence $[a,b] = [-2LU, 2LU]$, $b - a = 4LU$, and $11(b-a) = 44LU$. □

We now upper-bound $\mathbb{E}\mathrm{Rad}_{n,s}\big(\ell_{\mathcal{F}} - \ell_{f^*} : L^2 P((f-f^*)^2) \le r\big)$. What matters is that the bound takes advantage of the small variance, i.e., small values of r (it scales as $\sqrt{r}$). In Bartlett et al. (2005), such a bound is obtained by applying Bousquet's concentration inequality (Bousquet, 2002) directly to the localized class. Here we instead reuse Dudley's integral bound of Theorem 33 (recall $C = 8$ there), applied at the localized variance scale.

Theorem 58. *Suppose $\mathcal{F}$ is a VC class with parameter (v, A) and envelope U, that $f^* \in \mathcal{F}$, that ℓ satisfies (LIP), and that $n \geq 2$. If $\sqrt{rn} \geq 2UL\sqrt{v \log(2An)}$, then*

$$\mathbb{E}\mathrm{Rad}_{n,s}\big(\ell_{\mathcal{F}} - \ell_{f^*} \,:\, L^2 P((f-f^*)^2) \leq r\big) \leq 4(2+C^2)\sqrt{nrv\log(2An)}.$$

Proof. Since $\mathbb{E}\sum_i \eta_i \ell_{f^*}(X_i, Y_i) = 0$, the centering by ℓ_{f^*} does not change the expectation. Hence, with $\mathcal{F}_r := \{f \in \mathcal{F} : L^2 P((f-f^*)^2) \leq r\}$ and by contraction,

$$\mathbb{E}\mathrm{Rad}_{n,s}\big(\ell_{\mathcal{F}}-\ell_{f^*} \,:\, L^2 P((f-f^*)^2) \leq r\big) = \mathbb{E}\mathrm{Rad}_{n,s}(\ell_{\mathcal{F}_r}) \leq L\,\mathbb{E}\mathrm{Rad}_{n,s}(\mathcal{F}_r) = L\,\mathbb{E}\mathrm{Rad}_{n,s}(\mathcal{F}_r - f^*),$$

where the last equality uses again $\mathbb{E}\sum_i \eta_i f^*(X_i) = 0$. The class $\mathcal{F}_r - f^*$ is VC with parameters (v, A) and envelope $2U$, and $\sup_{h \in \mathcal{F}_r - f^*} P(h^2) \leq r/L^2$.

Case $r \leq 4U^2L^2$. Then $\sigma := \sqrt{r}/L \leq 2U$, and Theorem 33 applied with envelope $2U$ and $\sqrt{r}/L$ in place of $\sqrt{r}$ gives, since $2A(2U)/\sigma = 4AUL/\sqrt{r}$,

$$\mathbb{E}\mathrm{Rad}_{n,s}(\mathcal{F}_r - f^*) \leq \frac{8}{L}\sqrt{nrv\log\Big(\frac{4AUL}{\sqrt{r}}\Big)} + 8C^2Uv\log\Big(\frac{4AUL}{\sqrt{r}}\Big).$$

The assumption $\sqrt{rn} \geq 2UL\sqrt{v\log(2An)}$ implies $\sqrt{rn} \geq 2UL$ (as $A \geq e$ gives $\log(2An) \geq 1$, and $v \geq 1$), i.e., $2AUL/\sqrt{r} \leq A\sqrt{n}$ and therefore $4AUL/\sqrt{r} \leq 2An$, so both logarithms are at most $\log(2An)$. It also gives $2UL\,v\log(2An) \leq \sqrt{rn\,v\log(2An)}$. Therefore

$$L\,\mathbb{E}\mathrm{Rad}_{n,s}(\mathcal{F}_r - f^*) \leq 8\sqrt{nrv\log(2An)} + 4C^2\sqrt{nrv\log(2An)} = 4(2+C^2)\sqrt{nrv\log(2An)}.$$

Case $r > 4U^2L^2$. Then $\mathcal{F}_r = \mathcal{F}$ and every $h \in \mathcal{F} - f^*$ satisfies $P(h^2) \leq (2U)^2$, so Theorem 33 applies with envelope $2U$ and $\sqrt{r} = 2U$ (for which $2A(2U)/\sqrt{r} = 2A$):

$$L\,\mathbb{E}\mathrm{Rad}_{n,s}(\mathcal{F} - f^*) \leq 16\,UL\sqrt{nv\log(2A)} + 8C^2UL\,v\log(2A).$$

Since $\log(2A) \leq \log(2An)$; since $\sqrt{r} > 2UL$, the first term is at most $8\sqrt{nrv\log(2An)}$; and by $2UL\,v\log(2An) \leq \sqrt{rn\,v\log(2An)}$ the second is at most $4C^2\sqrt{nrv\log(2An)}$. The same bound follows. □

We are now able to state the main result of the section. It provides a fast rate for the excess risk: it scales as $1/n$, which is better than the standard rate $1/\sqrt{n}$ discussed at the beginning of the section.

Theorem 59. *Suppose $\mathcal{F}$ is a VC class with parameter (v, A) and envelope U, $f^* \in \mathcal{F}$, $n \geq 2$, and the loss is L-Lipschitz (LIP) and B-convex (CONV). Then, with probability $1-\delta$,*

$$P(\ell_{\hat{f}} - \ell_{f^*}) \leq (C_1 \vee C_2)\Big(BL^2 \vee \frac{U^2}{B} \vee LU\Big)\frac{v\log(2An/\delta)}{n},$$

where $C_1 := 32c_1(2+C^2)^2$ and $C_2 := 44 + 2c_2$.

Proof. Let $\theta_0 \geq 1$ be specified below. Multiplying the conclusion of Theorem 58 by B_L/n and inflating the bound by the factor $\theta_0 \geq 1$, we obtain that, whenever $\sqrt{rn} \geq 2UL\sqrt{v\log(2An)}$,

$$\frac{B_L}{n}\mathbb{E}\mathrm{Rad}_{n,s}\big(\ell_{\mathcal{F}}-\ell_{f^*} \,:\, L^2P((f-f^*)^2) \leq r\big) \leq \psi(r), \qquad \psi(r) := 4\theta_0(2+C^2)B_L\sqrt{n^{-1}rv\log(2An)}.$$

The function ψ is sub-root, and its fixed point $r^* = \psi(r^*)$ is

$$r^* = \{4\theta_0(2+C^2)B_L\}^2 \, v n^{-1} \log(2An).$$

By Theorem 57, with probability $1-\delta$,

$$P(\ell_{\hat{f}} - \ell_{f^*}) \le \frac{2c_1 r^*}{B_L} + (44LU + 2c_2 B_L)\, n^{-1} \log\left(\frac{1}{\delta}\right), \tag{2}$$

provided that ψ dominates the Rademacher term for *all* $r \ge r^*$. By Theorem 58, this holds as soon as $\sqrt{rn} \ge 2UL\sqrt{v\log(2An)}$ for all $r \ge r^*$, i.e., as soon as $\sqrt{r^* n} \ge 2UL\sqrt{v\log(2An)}$, i.e.,

$$4\theta_0(2+C^2)B_L \ge 2UL, \qquad \text{i.e.} \qquad \theta_0 \ge \frac{UL}{(2+C^2)B_L}.$$

We therefore set $\theta_0 := \dfrac{UL}{(2+C^2)B_L} \vee 1$, so that $4\theta_0(2+C^2)B_L = 4UL \vee 4(2+C^2)B_L$ and

$$r^* = (4UL \vee 4(2+C^2)B_L)^2 v n^{-1}\log(2An) = \Big\{16U^2L^2 \vee 16(2+C^2)^2 B_L^2\Big\}\frac{v\log(2An)}{n}.$$

For the first term of (2), using $B_L = BL^2$, so that $U^2L^2/B_L = U^2/B$,

$$\frac{2c_1}{B_L} r^* = 2c_1\Big\{16\,\frac{U^2}{B} \;\vee\; 16(2+C^2)^2 BL^2\Big\}\frac{v\log(2An)}{n} \;\le\; 32c_1(2+C^2)^2\Big(BL^2 \vee \frac{U^2}{B}\Big)\frac{v\log(2An)}{n},$$

where we used $1 \le (1+C^2)^2$. This is $C_1\big(BL^2 \vee U^2/B\big)\, n^{-1} v\log(An)$. For the second term of (2), since $44LU + 2c_2B_L \le (44+2c_2)(LU \vee B_L) = C_2(LU \vee BL^2)$, it is at most $C_2\,(LU \vee BL^2)\, n^{-1}\log(1/\delta)$. To finish, let $M := BL^2 \vee U^2/B \vee LU$ and use $v \ge 1$ and $\log(2An) + \log(1/\delta) = \log(2An/\delta)$, to get that with probability $1-\delta$,

$$P(\ell_{\hat{f}} - \ell_{f^*}) \le (C_1 \vee C_2)\, M\, \frac{v\log(2An) + \log(1/\delta)}{n} \le (C_1 \vee C_2)\, M\, \frac{v\log(2An/\delta)}{n}.$$

□

9 Auxiliary results

Lemma 60. *Under (Reg2), we have*

$$\|f(x) - f(y) - (x-y)^T \nabla f(y)\| \le \frac{1}{2} L \|x-y\|^2$$

Proof. Consider the function $F(t) = f(tx + (1-t)y)$, $t \in [0,1]$. We have that

$$f(x) - f(y) = \int_0^1 F'(t)\mathrm{d}t = (y-x)^T \int_0^1 \nabla f(tx + (1-t)y)\mathrm{d}t$$

Consequently

$$f(x) - f(y) - (x-y)^T\nabla f(y) = (y-x)^T \int_0^1 (\nabla f(tx+(1-t)y) - \nabla f(y))\mathrm{d}t$$

and we conclude using the L-smoothness. □

Lemma 61. *Let $b > 0$ and $c > 0$. Suppose that $z^2 \le bz + c$, then $z \le b + \sqrt{c}$.*

Proof. Solve $z^2 - bz - c = 0$. There is two roots and only the positive one, $1/2(b + \sqrt{b^2+4c})$ is of interest. We have that $z \le 1/2(b + \sqrt{b^2+4c}) \le b + \sqrt{c}$. □

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